



Camera Module Installation Guide for AAEON UP Boards

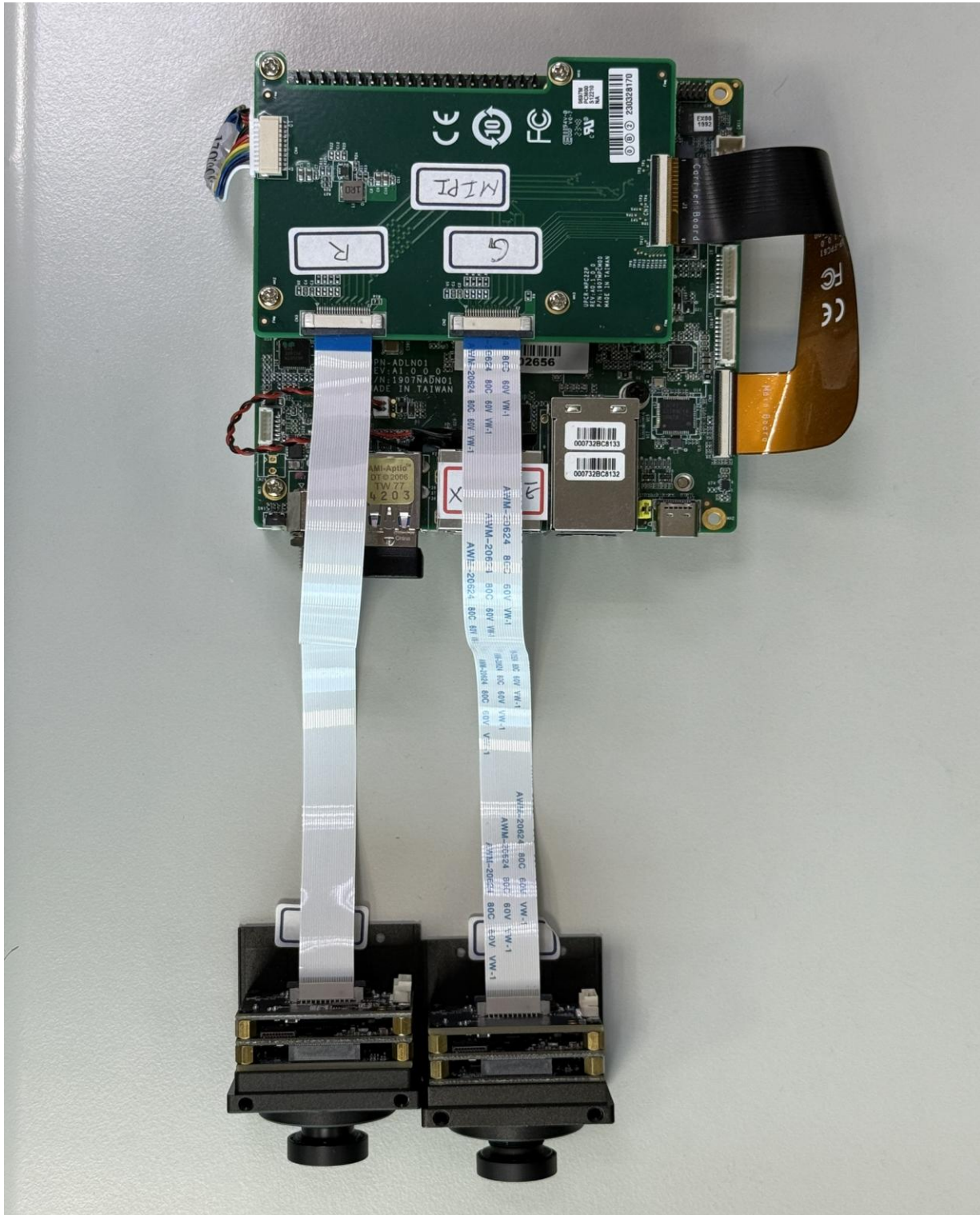
V1.0

Hardware Setup

■ GMSL Camera x 8



■ MIPI Camera x 2



Release Package

- Release package will include files as below. Provided Package is verified on Ubuntu 24.04 and Ubuntu 22.04.

```
-release_package
  ApproCam_Installation_Guide_AAEON_UP_v1.0.pdf
  -kernel_image
    linux-headers-6.6.65--000_6.6.65-0_amd64.deb
    linux-image-6.6.65--000_6.6.65-0_amd64.deb
    linux-libc-dev_6.6.65-0_amd64.deb
  -ubuntu
    ipu_install_22.04.tgz
    ipu_install_24.04.tgz
  -test_cmd
    adl_gmsl_up_abcdefgh.sh
    adl_mipi_up_bc.sh
    camenv.sh
    mtl_gmsl_up_abcdefgh.sh
    mtl_mipi_up_ae.sh
    stopDemo.sh
```

Ubuntu Setup

■ Software Package Install

1. Ubuntu Software Package Update

```
$ sudo apt update
$ sudo apt upgrade
$ export DEBIAN_FRONTEND=noninteractive
$ sudo apt install vim ocl-icd-libopencl1 curl openssl-server net-tools gir1.2-gst-plugins-bad-1.0 gir1.2-gst-plugins-base-1.0 gir1.2-gstreamer-1.0 gir1.2-gst-rtsp-server-1.0 gstreamer1.0-alsa gstreamer1.0-gl gstreamer1.0-gtk3 gstreamer1.0-opencv gstreamer1.0-plugins-bad gstreamer1.0-plugins-bad-apps gstreamer1.0-plugins-base gstreamer1.0-plugins-base-apps gstreamer1.0-plugins-good gstreamer1.0-plugins-ugly gstreamer1.0-pulseaudio gstreamer1.0-qt5 gstreamer1.0-rtsp gstreamer1.0-tools gstreamer1.0-x intel-media-va-driver-non-free libdrm-amdgpu1 libdrm-common libdrm-dev libdrm-intel1 libdrm-nouveau2 libdrm-radeon1 libdrm-tests libdrm2 libgstrtspserver-1.0-dev libgstrtspserver-1.0-0 libgstreamer-gl1.0-0 libgstreamer-opencv1.0-0 libgstreamer-plugins-bad1.0-0 libgstreamer-plugins-bad1.0-dev libgstreamer-plugins-base1.0-0 libgstreamer-plugins-base1.0-dev libgstreamer-plugins-good1.0-0 libgstreamer-plugins-good1.0-dev libgstreamer1.0-0 libgstreamer1.0-dev libigdgmm-dev libigdgmm12 libigfxcmrt-dev libigfxcmrt7 libmfx-gen1.2 libtpms-dev libtpms0 libva-dev libva-drm2 libva-glx2 libva-wayland2 libva-x11-2 libva2 libwayland-bin libwayland-client0 libwayland-cursor0 libwayland-dev libwayland-doc libwayland-egl-backend-dev libwayland-egl1 libwayland-server0 libxatracker2 linux-firmware mesa-utils mesa-va-drivers mesa-udpau-drivers mesa-vulkan-drivers libvpl-dev libmfx-gen-dev ovmf ovmf-ia32 qemu-block-extra qemu-guest-agent qemu-system qemu-system-arm qemu-system-common qemu-system-data qemu-system-gui qemu-system-mips qemu-system-misc qemu-system-ppc qemu-system-s390x qemu-system-sparc qemu-system-x86 qemu-user qemu-user-binfmt qemu-utils va-driver-all vainfo weston xserver-xorg-core xserver-common xnest xserver-xorg-dev xvfb libvirt0 libvirt-clients libvirt-daemon libvirt-daemon-config-network libvirt-daemon-config-nwfilter libvirt-daemon-driver-lxc libvirt-daemon-driver-qemu libvirt-daemon-driver-storage-gluster libvirt-daemon-driver-storage-iscsi-direct libvirt-daemon-driver-storage-rbd libvirt-daemon-driver-storage-zfs libvirt-daemon-driver-vbox libvirt-daemon-driver-xen libvirt-daemon-system libvirt-daemon-system-systemd libvirt-dev libvirt-doc libvirt-login-shell libvirt-sanlock libvirt-wireshark libnss-libvirt swtpm swtpm-tools bmap-tools adb autoconf automake libtool cmake g++ gcc git intel-gpu-tools libssl3 libssl-dev make mosquitto mosquitto-clients build-essential apt-transport-https default-jre docker-compose ffmpeg git-lfs gnuplot lzip2 libglew-dev libglm-dev libstdc++2-dev mc openssl pciutils python3-pandas python3-pip python3-seaborn terminator vim wmcctl wayland-protocols gdbserver ethtool iperf3 msr-tools powertop linuxptp lsscsi tpm2-tools tpm2-abrmd binutils cifs-utils i2c-tools xdotool gnupg lsb-release intel-opencl-icd ethtool iproute2 socat virt-viewer spice-client-gtk dbus-x11 -y --allow-downgrades
```

2. Intel IPU Package Install

- Apply IPU package in release package

- ◆ **Ubuntu 22.04**

```
$ cd release_package/ubuntu/  
$ tar xvfz ipu_install_22.04.tgz  
$ cd ipu_install/  
$ sudo cp -r ipu6-camera-bins/include/* /usr/include/  
$ sudo cp -r ipu6-camera-bins/lib/* /usr/lib/  
$ sudo cp -r ./out/install/etc/* /etc/  
$ sudo cp -r ./out/install/include/* /usr/include/  
$ sudo cp -r ./out/install/lib/* /usr/lib/
```

- ◆ **Ubuntu 24.04**

```
$ cd release_package/ubuntu/  
$ tar xvfz ipu_install_24.04.tgz  
$ cd ipu_install/  
$ sudo cp -r ipu6-camera-bins/include/* /usr/include/  
$ sudo cp -r ipu6-camera-bins/lib/* /usr/lib/  
$ sudo cp -r ./out/install/etc/* /etc/  
$ sudo cp -r ./out/install/include/* /usr/include/  
$ sudo cp -r ./out/install/lib/* /usr/lib/  
$ sudo cp -r ./out/install/usr/* /usr/
```

- Override IPU frequency

- ◆ `$ sudo vi /etc/modprobe.d/ipu.conf`

```
options intel-ipu6 isys_freq_override=500
```

- Add execute attribute for the test command

```
$ cd release_package/ubuntu/test_cmd/  
$ chmod +x *.sh
```

Kernel update

1. Install Kernel Image in release package

```
$ cd release_package/kernel_image/  
$ sudo dpkg -i linux-image-6.6.65--000_6.6.65-0_amd64.deb  
$ sudo dpkg -i linux-headers-6.6.65--000_6.6.65-0_amd64.deb  
$ sudo dpkg -i linux-libc-dev_6.6.65-0_amd64.deb
```

2. Modify Grub Configuration for New Kernel Image

■ \$ sudo vim /etc/default/grub

```
#GRUB_DEFAULT=0  
GRUB_DEFAULT="Advanced options for Ubuntu>Ubuntu, with Linux 6.6.65--000"  
GRUB_TIMEOUT_STYLE=hidden  
GRUB_TIMEOUT=0  
GRUB_DISTRIBUTOR=`(. /etc/os-release; echo ${NAME:-Ubuntu}) 2>/dev/null || echo Ubuntu`  
#GRUB_CMDLINE_LINUX_DEFAULT="quiet splash"  
GRUB_CMDLINE_LINUX_DEFAULT="noquiet nosplash i915.force_probe=* i915.enable_guc=3  
i915.max_vfs=7 udmabuf.list_limit=8192"  
GRUB_CMDLINE_LINUX=""
```

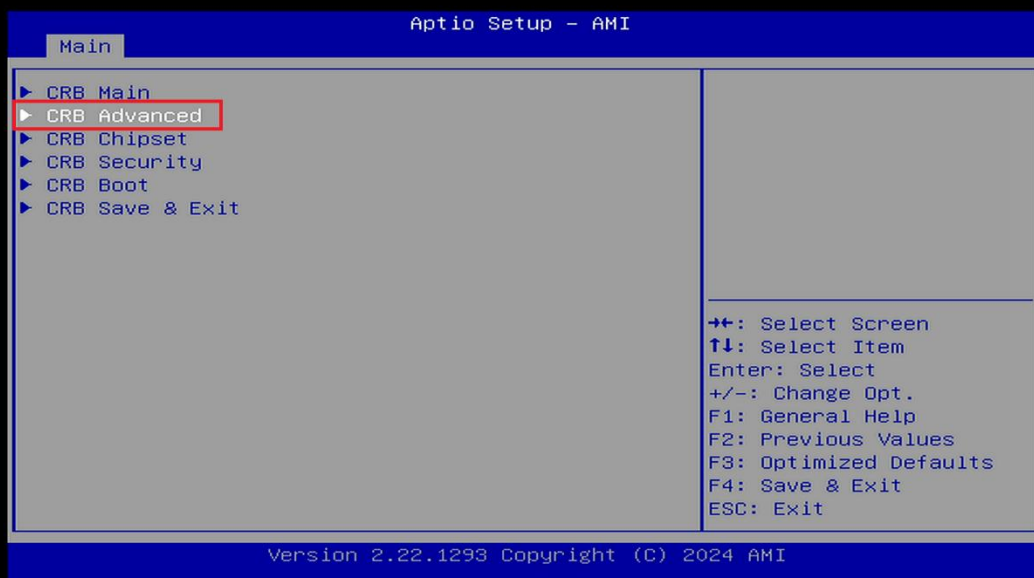
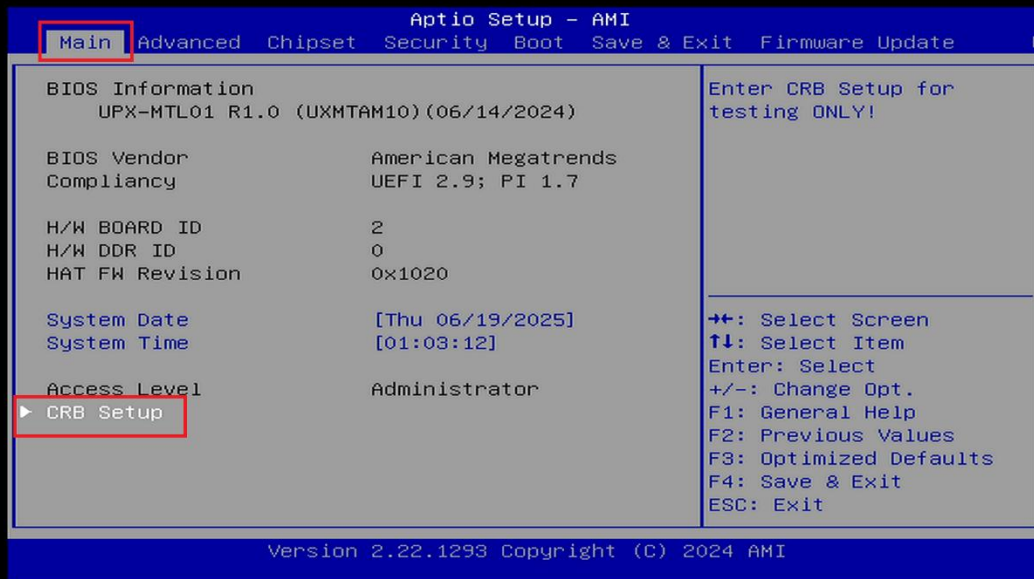
■ Update Grub configuration and reboot for applying new Kernel Image

```
$ sudo update-grub  
$ sudo reboot
```

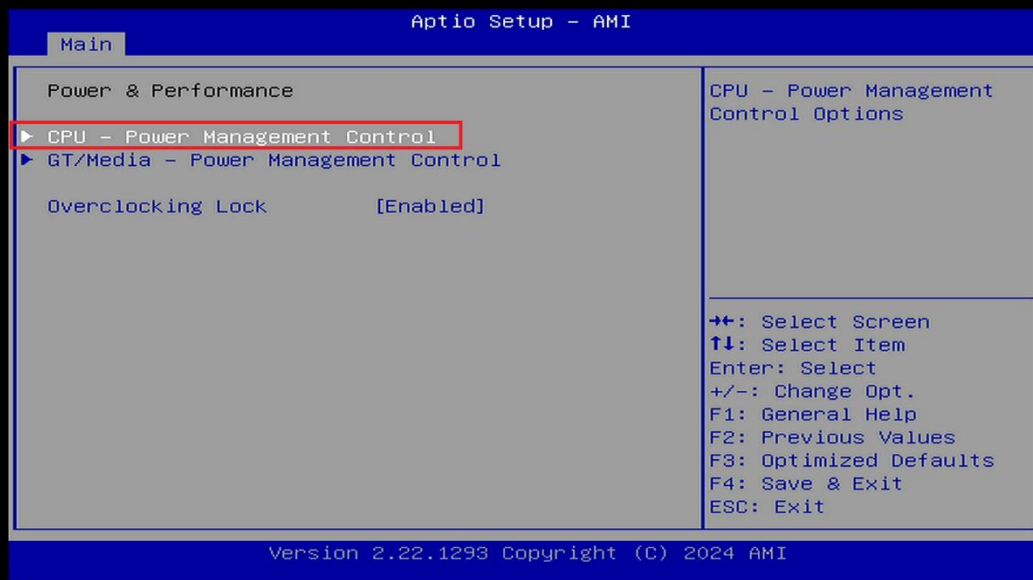
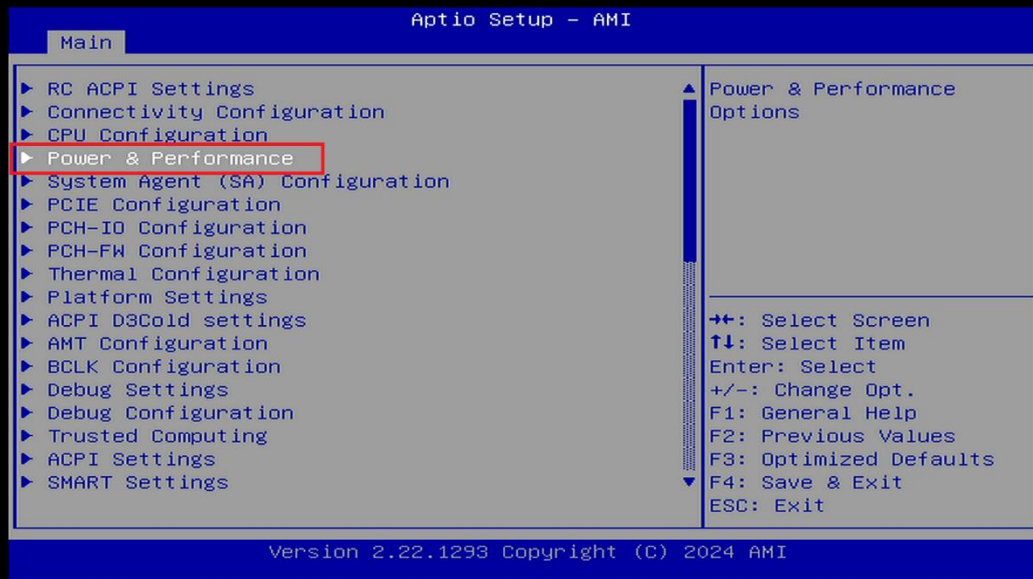
BIOS Setup (Meteor Lake)

MISC. Setting

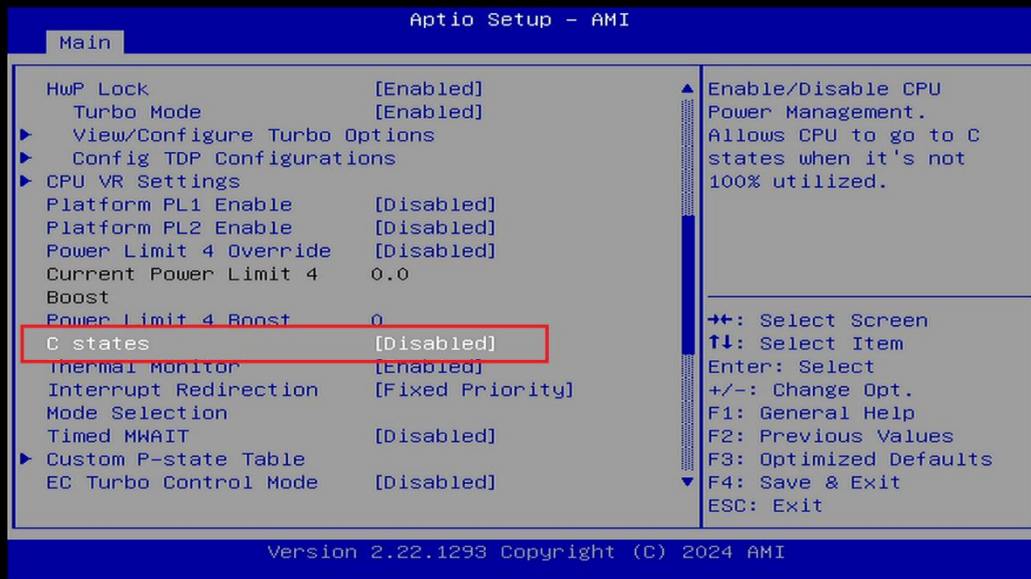
■ Disable C-states (step 1) - (Meteor Lake)



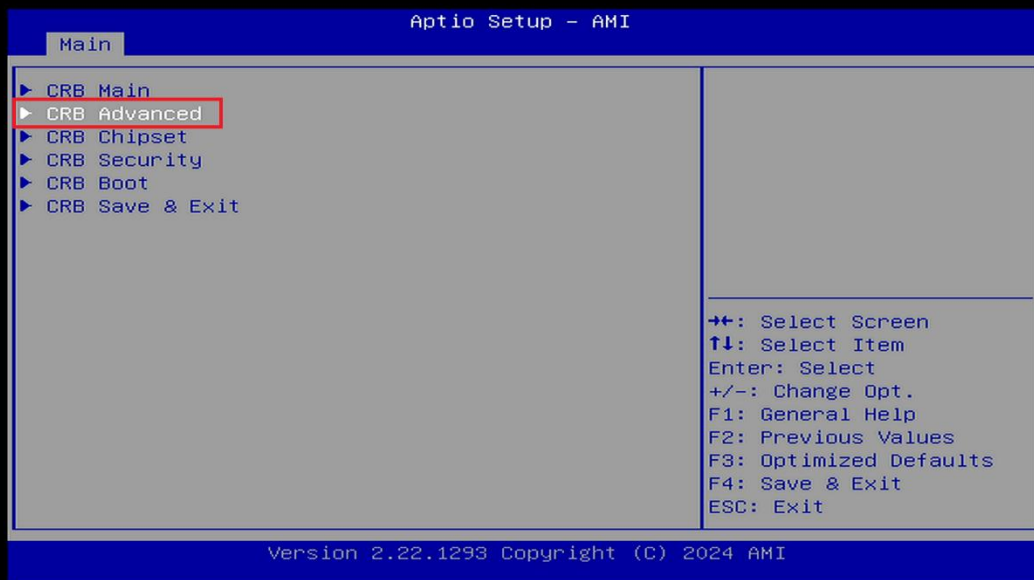
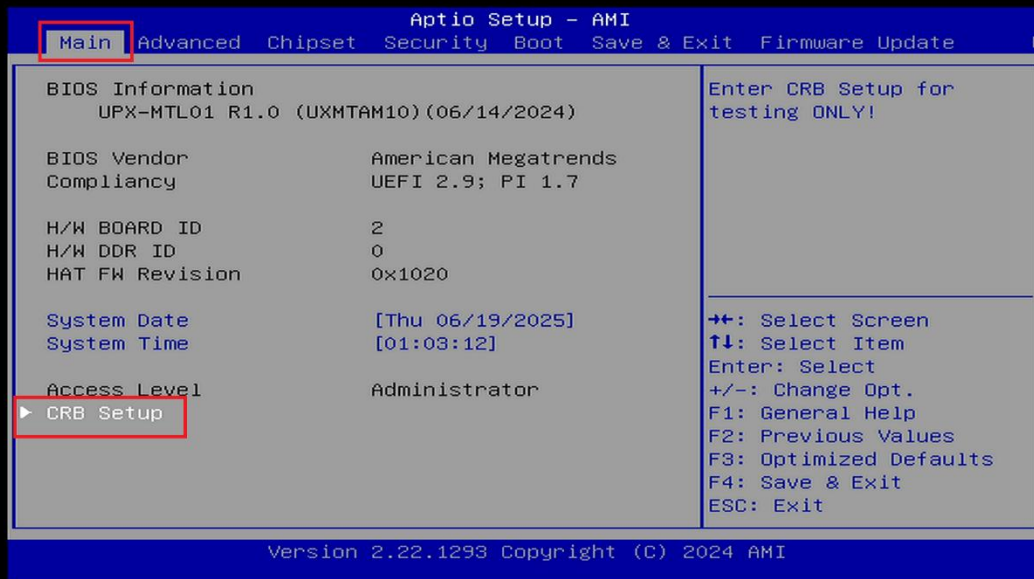
■ Disable C-states (step 2) - (Meteor Lake)



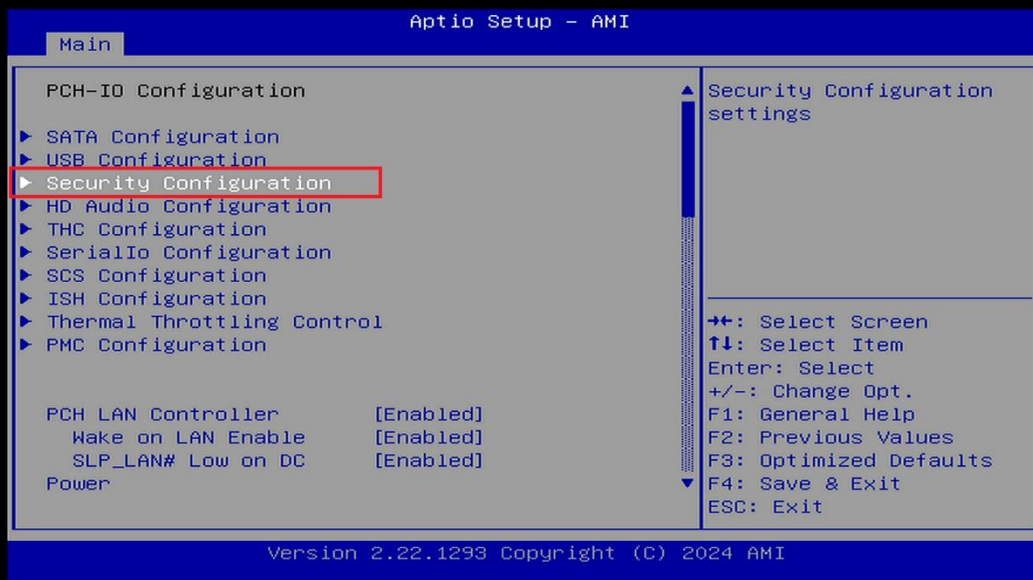
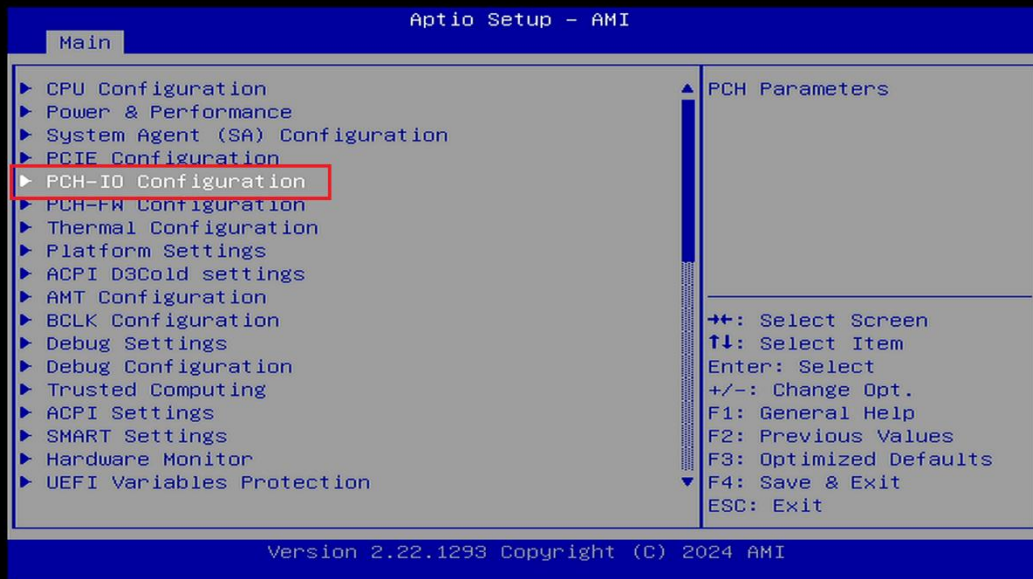
■ Disable C-states (step 3) - (Meteor Lake)



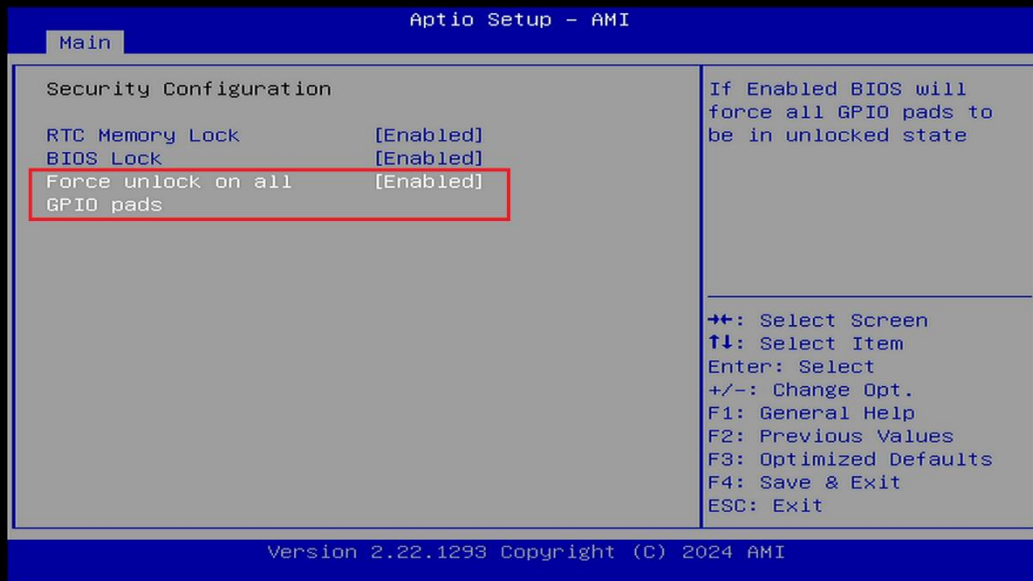
■ Force unlock on all GPIO pads (step 1) - (Meteor Lake)



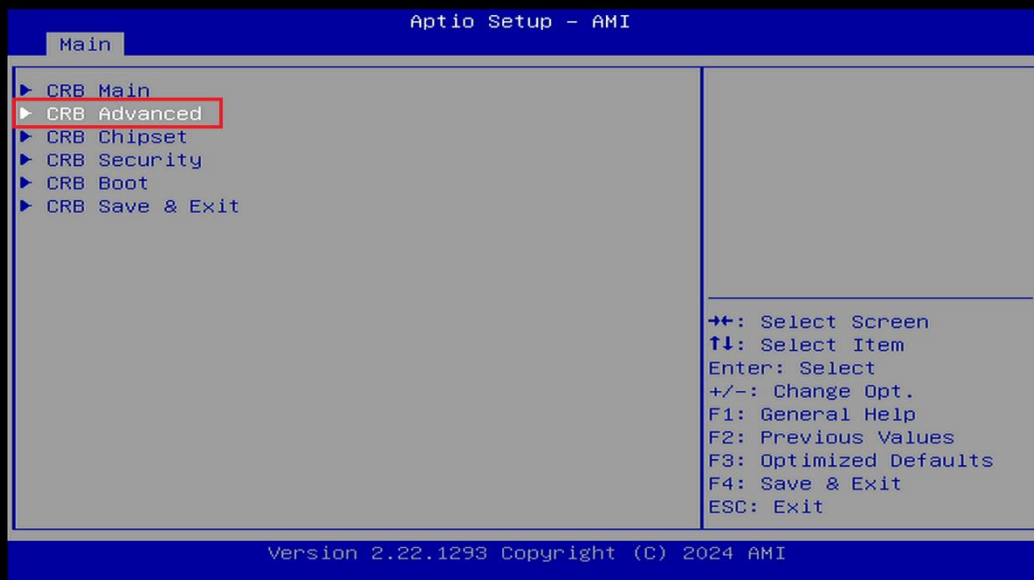
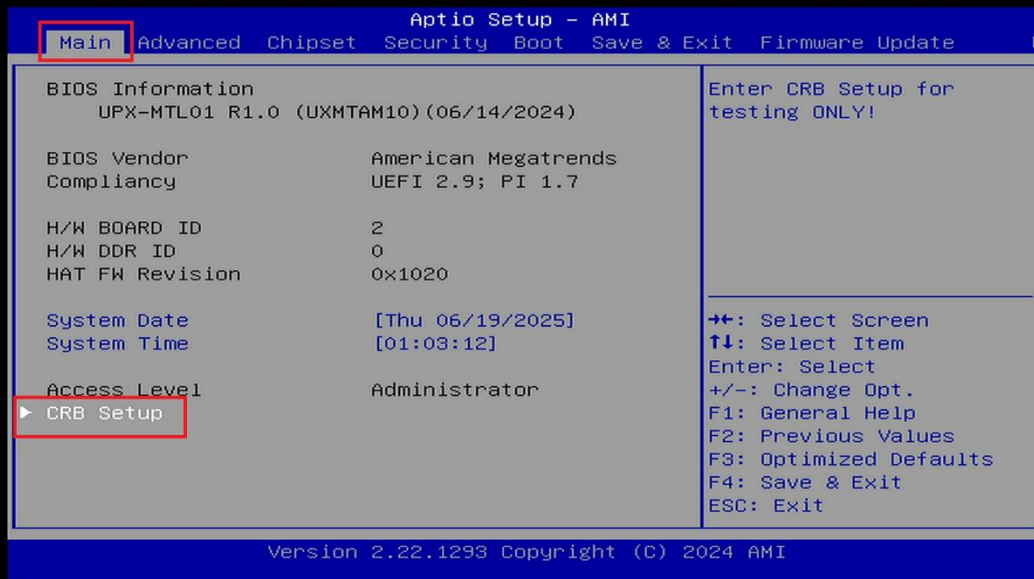
■ Force unlock on all GPIO pads (step 2) - (Meteor Lake)



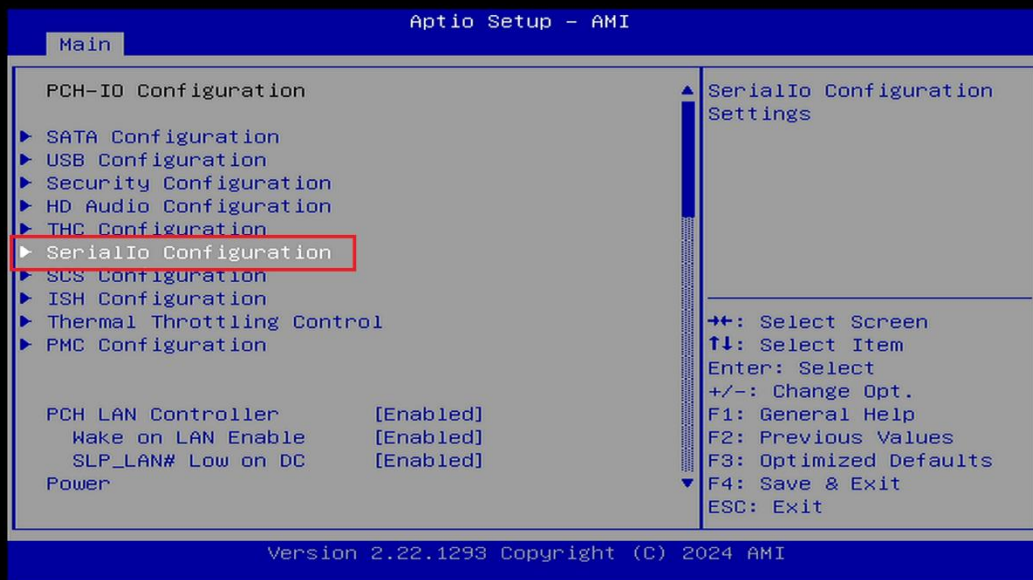
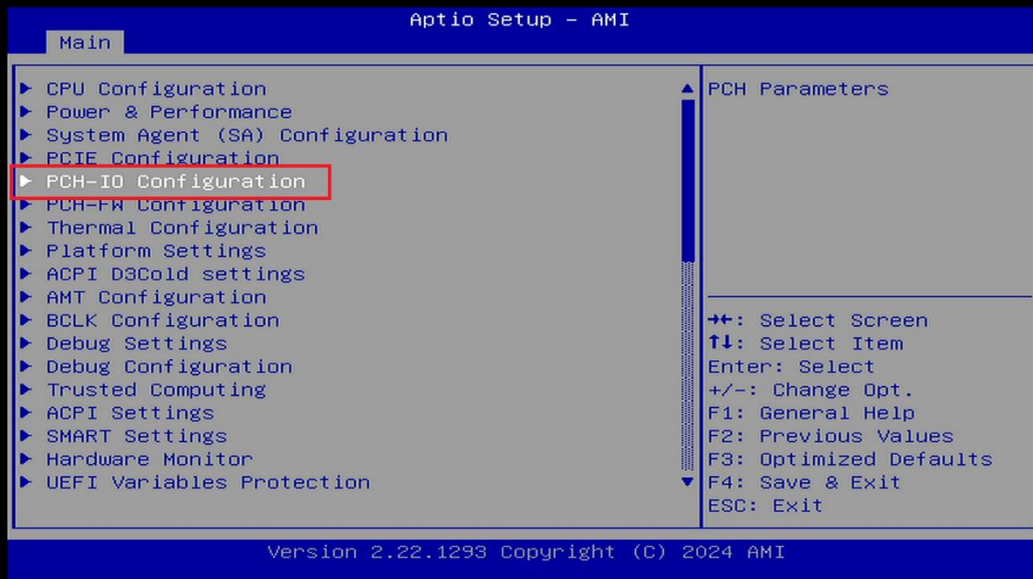
■ Force unlock on all GPIO pads (step 3) - (Meteor Lake)



■ I2C Enable (step 1) - (Meteor Lake)

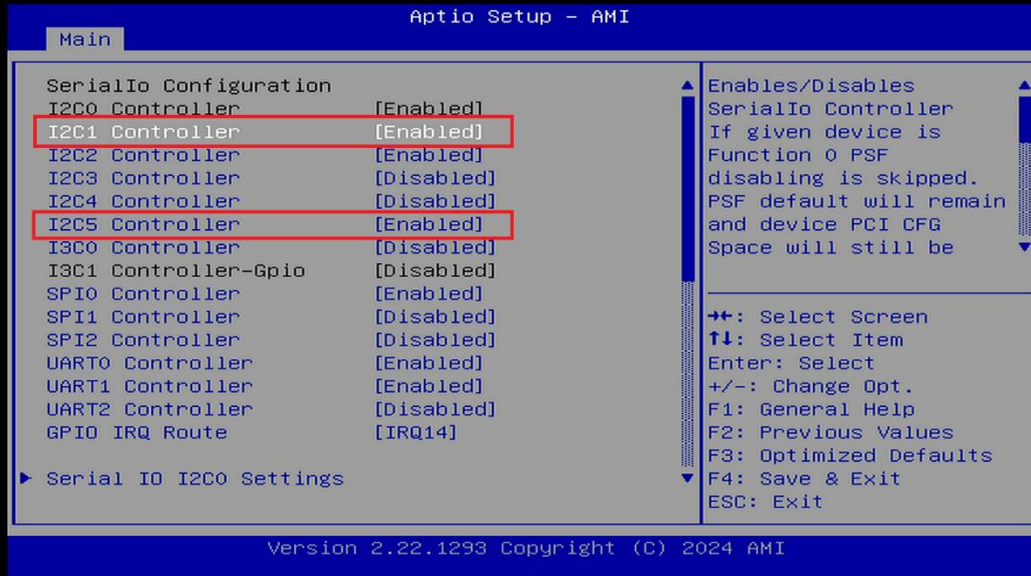


■ I2C Enable (step 2) - (Meteor Lake)

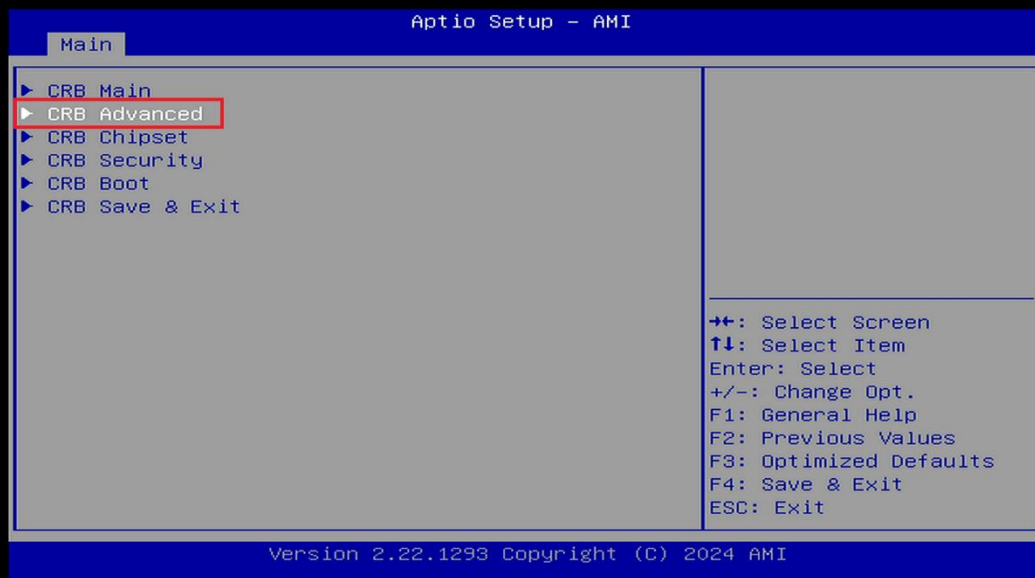
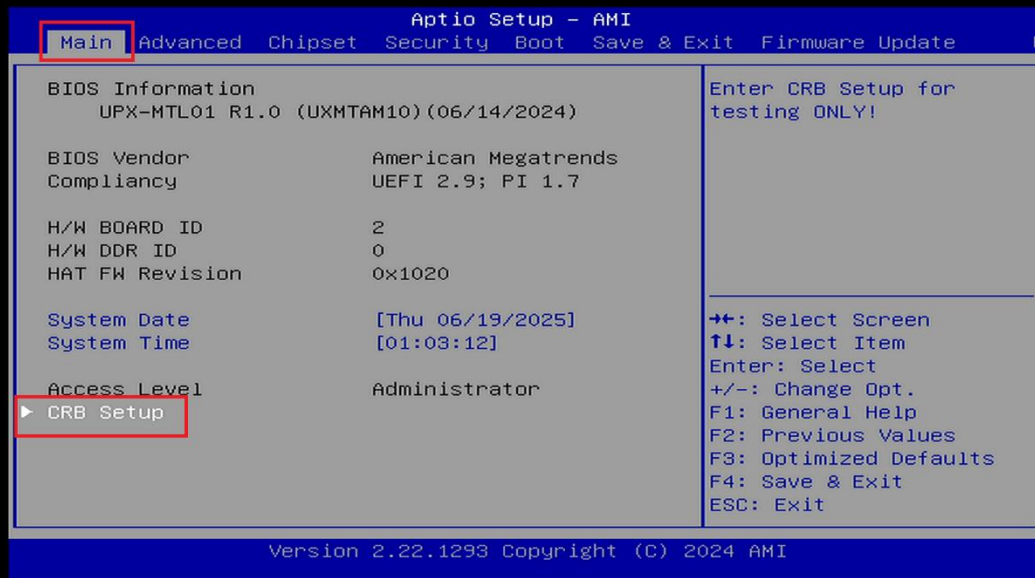


■ I2C Enable (step 3) - (Meteor Lake)

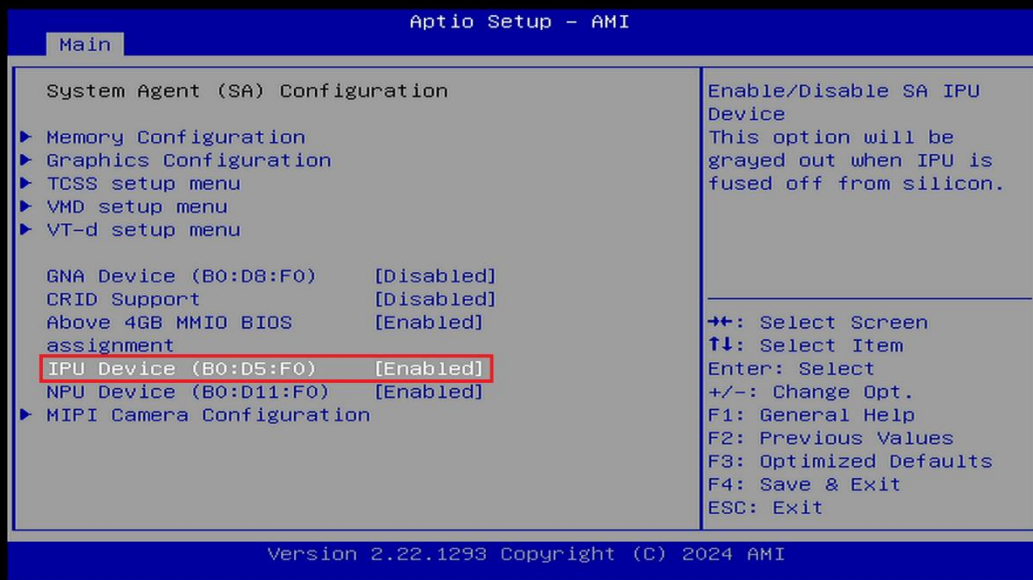
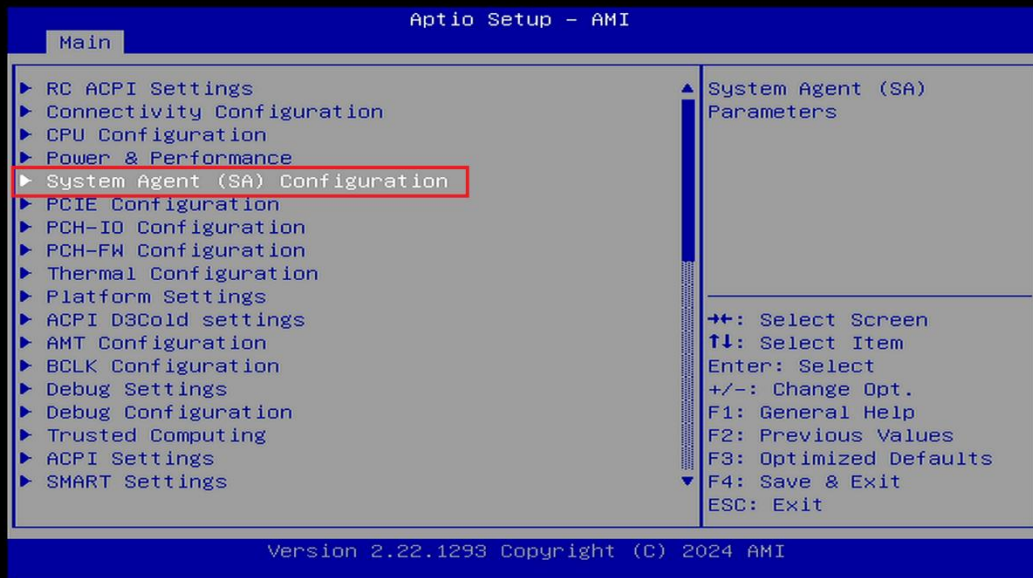
I2C1 and I2C5 must enable.



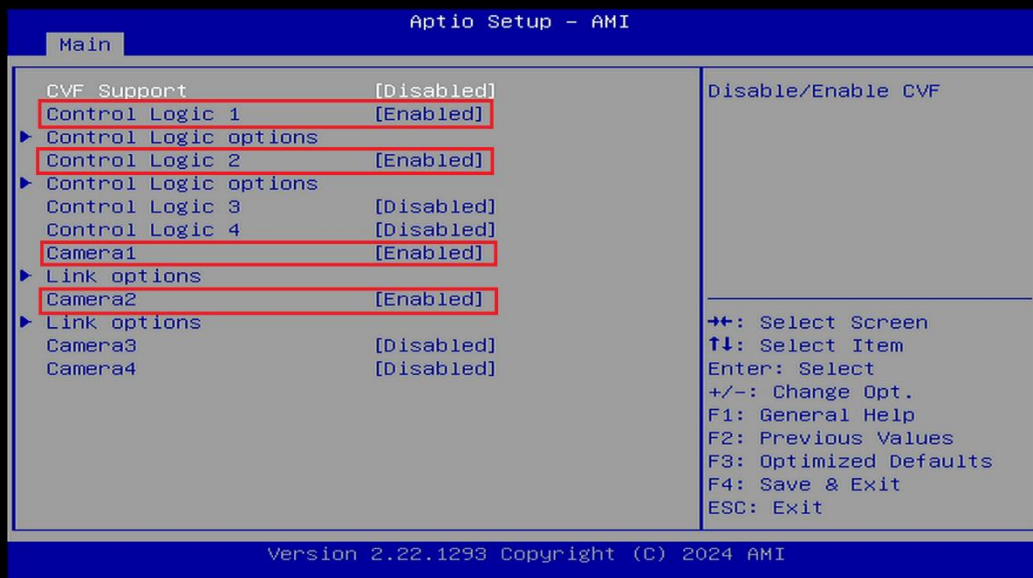
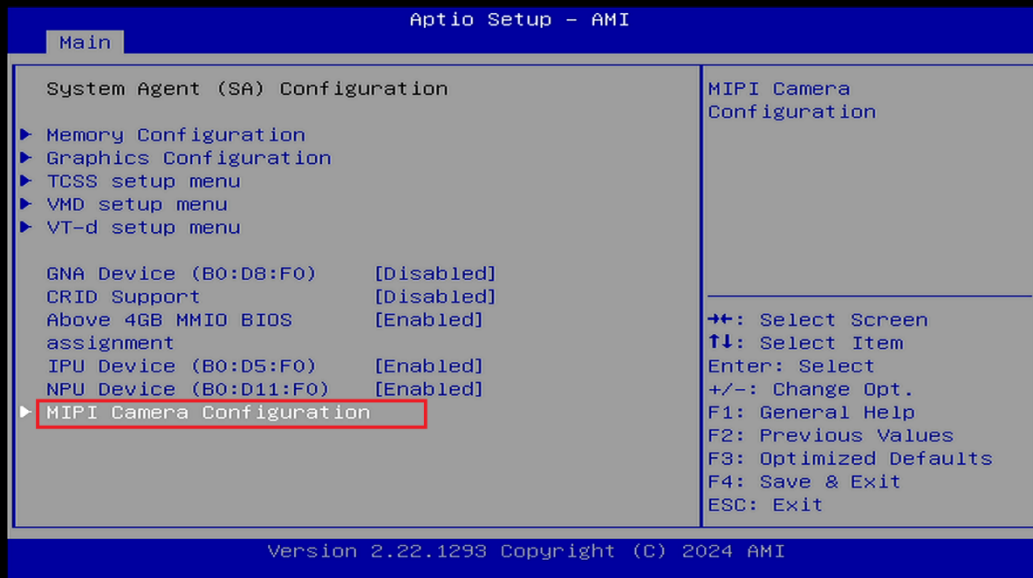
GMSL Camera x 8 (step 1)- (Meteor Lake)



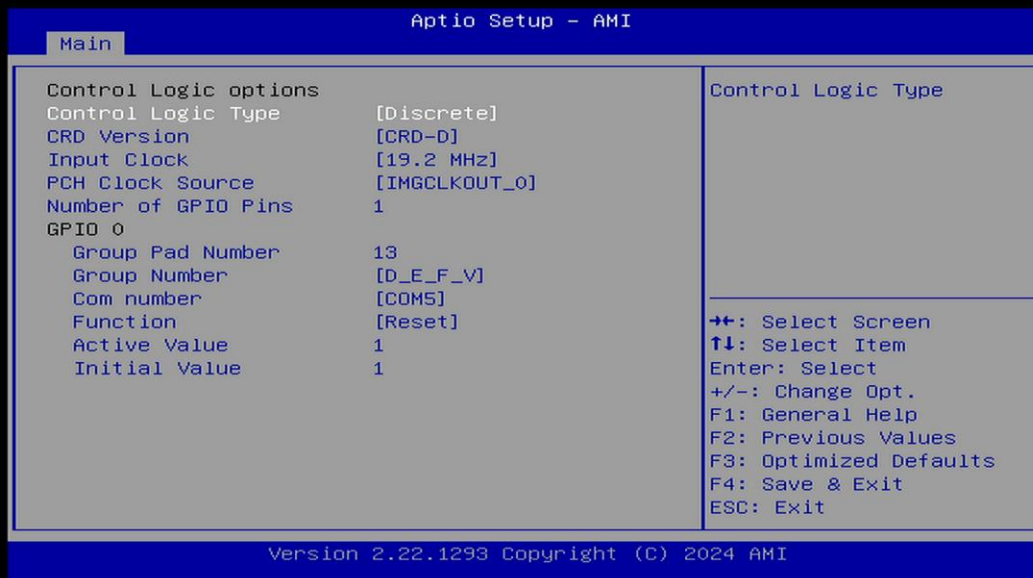
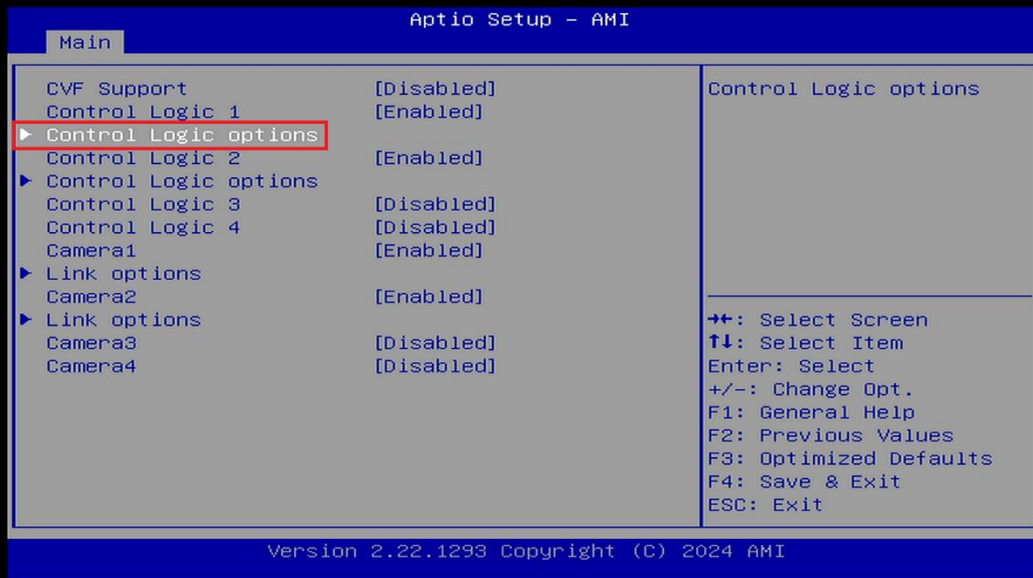
GMSL Camera x 8 (step 2)- (Meteor Lake)



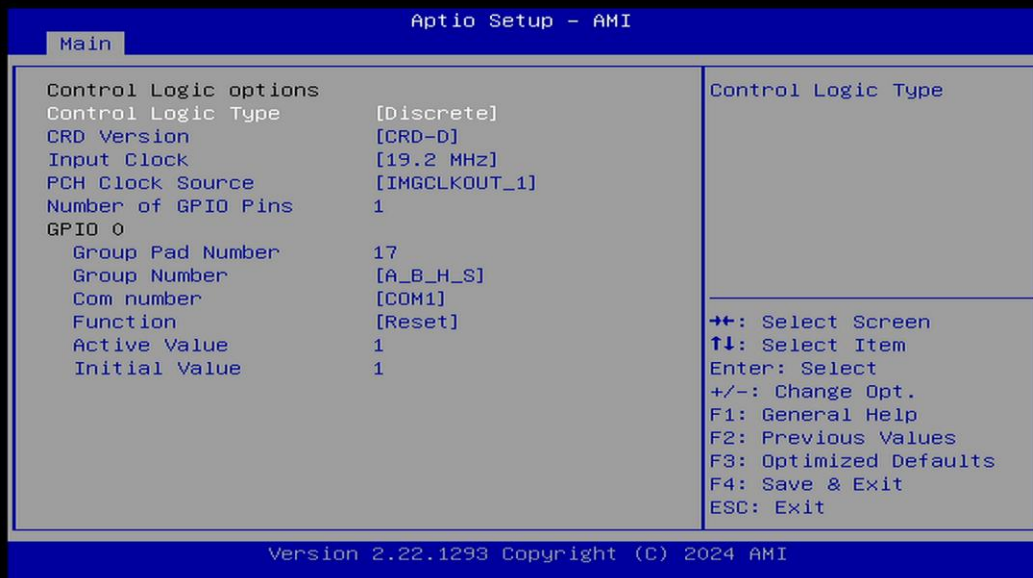
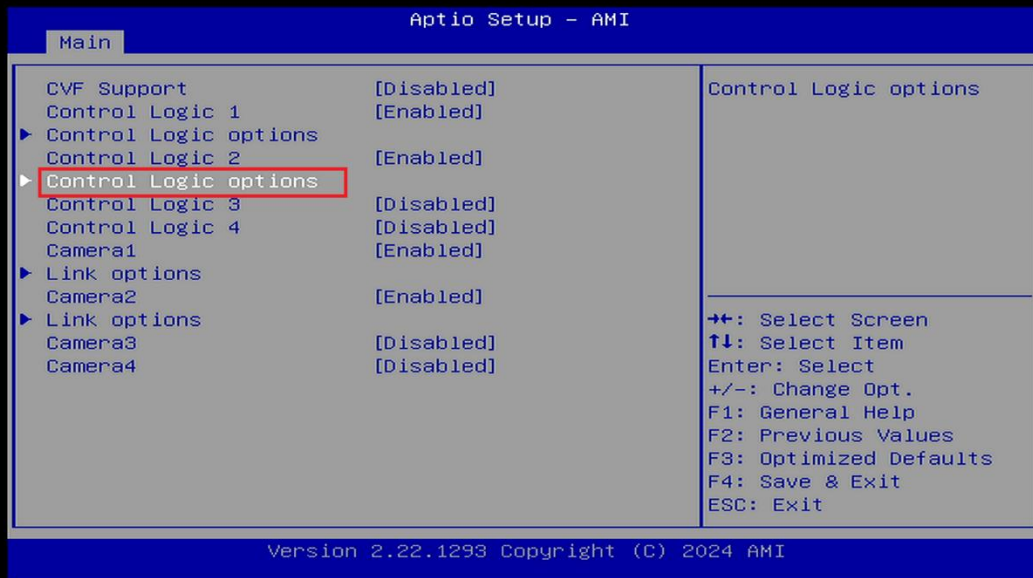
GMSL Camera x 8 (step 3)- (Meteor Lake)



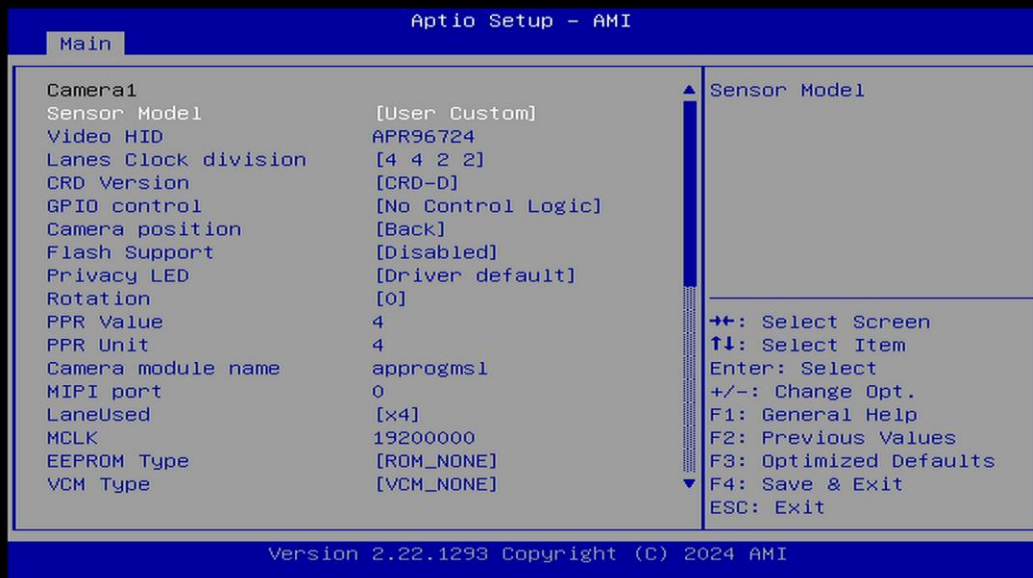
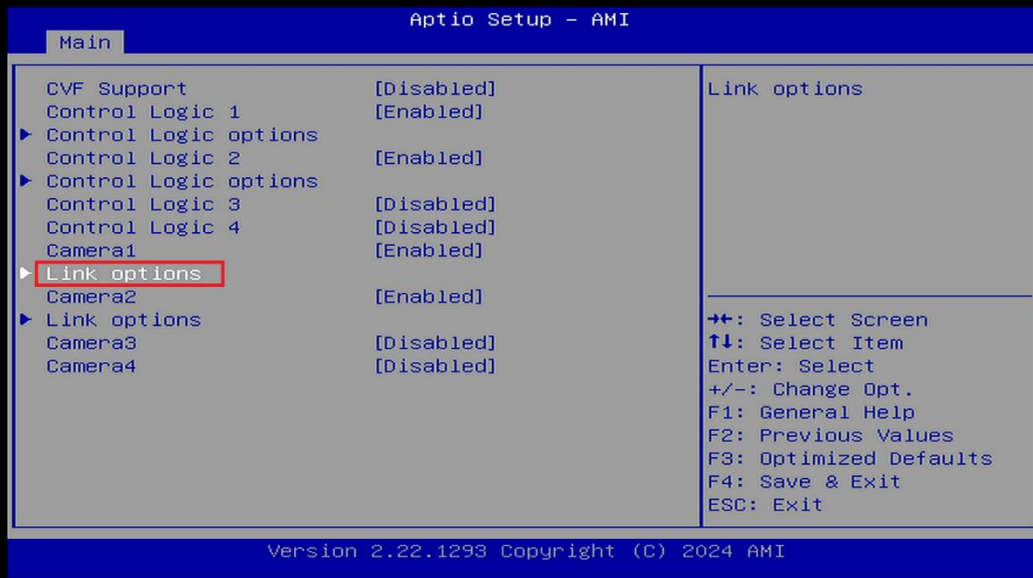
GMSL Camera x 8 (step 4)- (Meteor Lake)



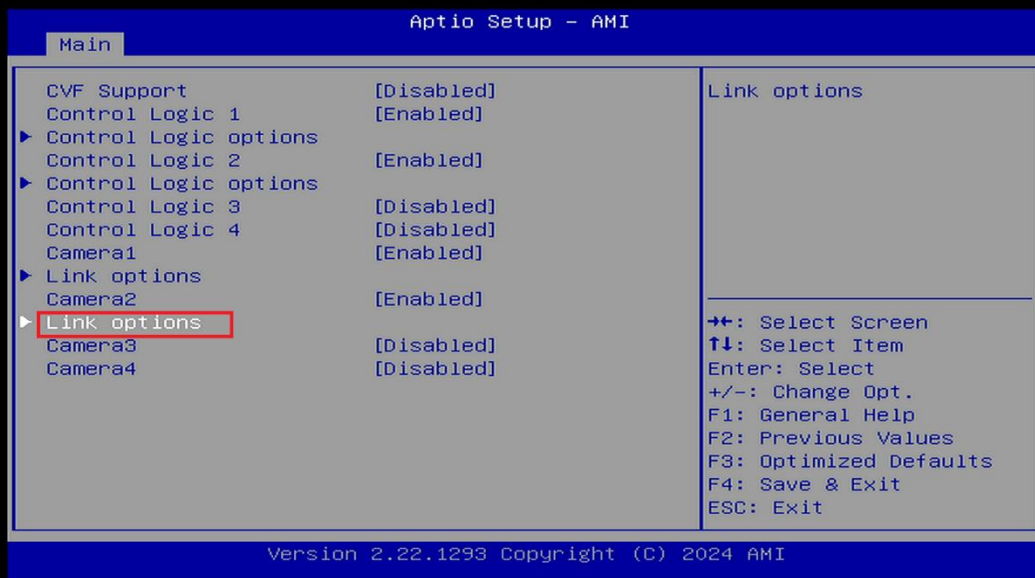
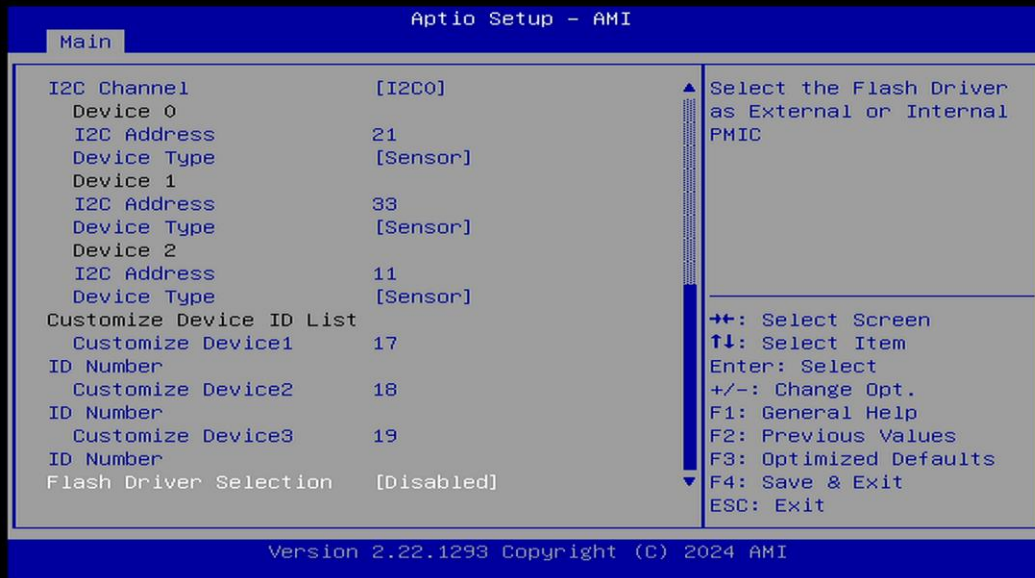
GMSL Camera x 8 (step 5)- (Meteor Lake)



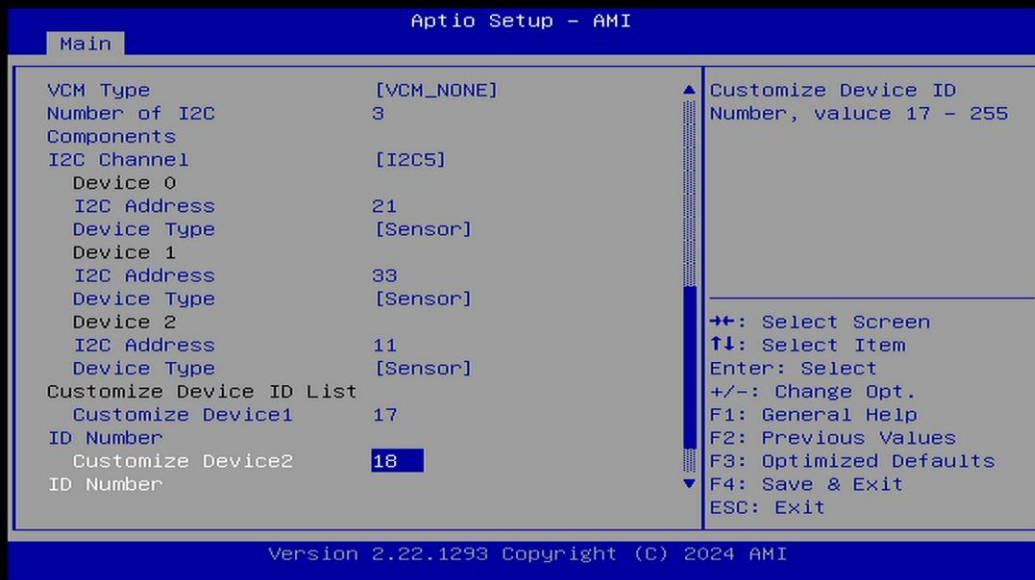
GMSL Camera x 8 (step 6)- (Meteor Lake)



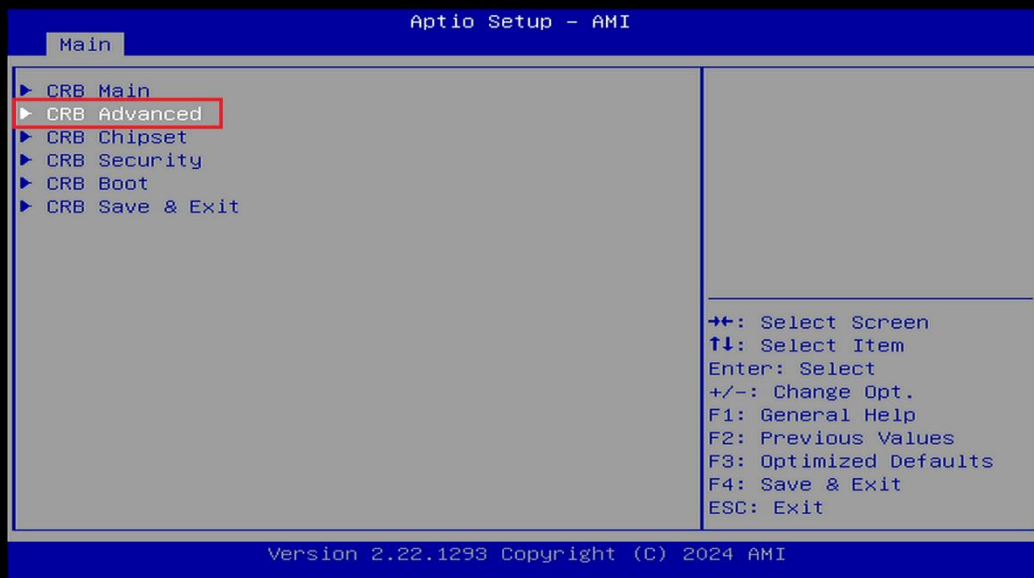
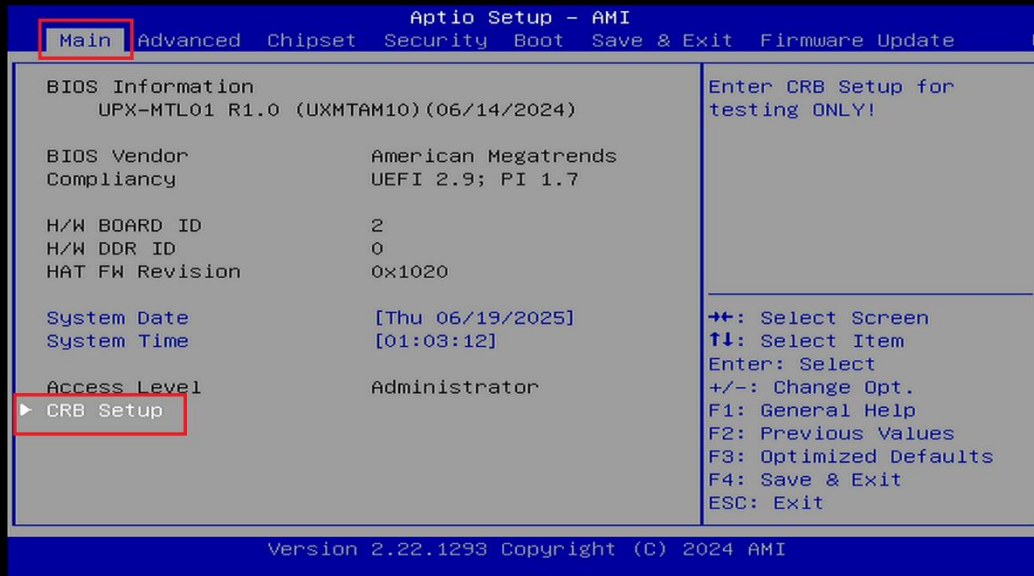
GMSL Camera x 8 (step 7)- (Meteor Lake)



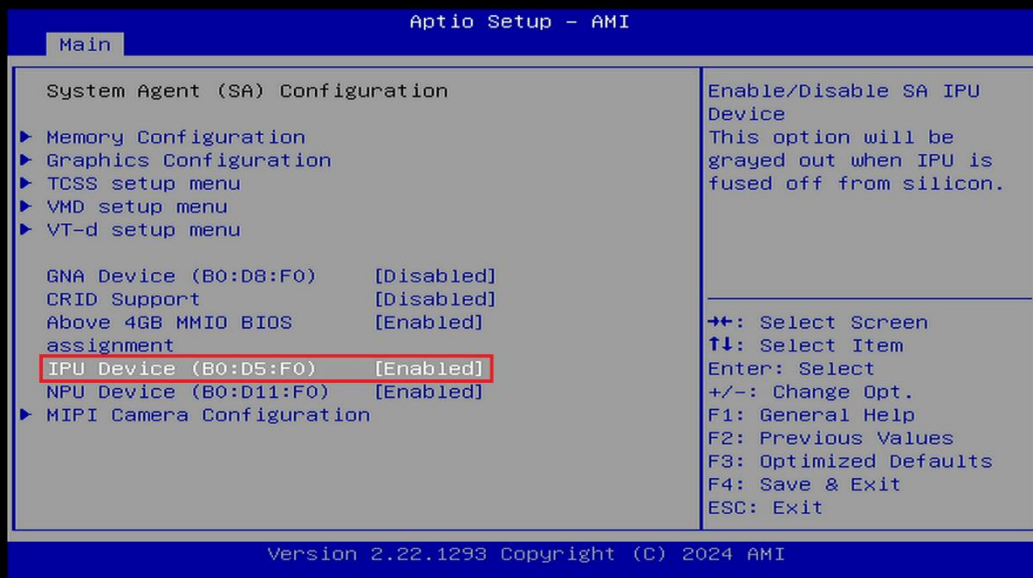
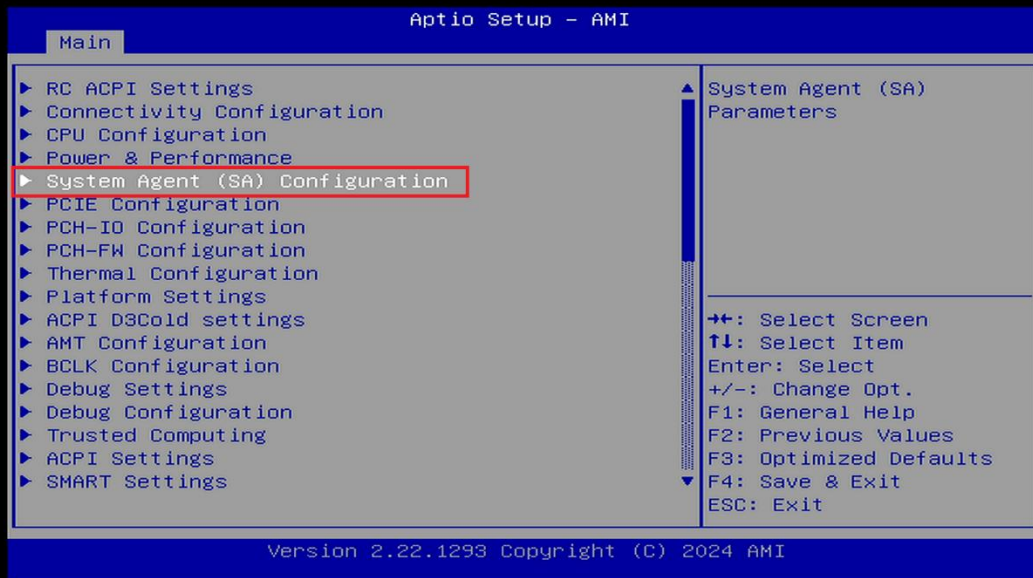
GMSL Camera x 8 (step 8)- (Meteor Lake)



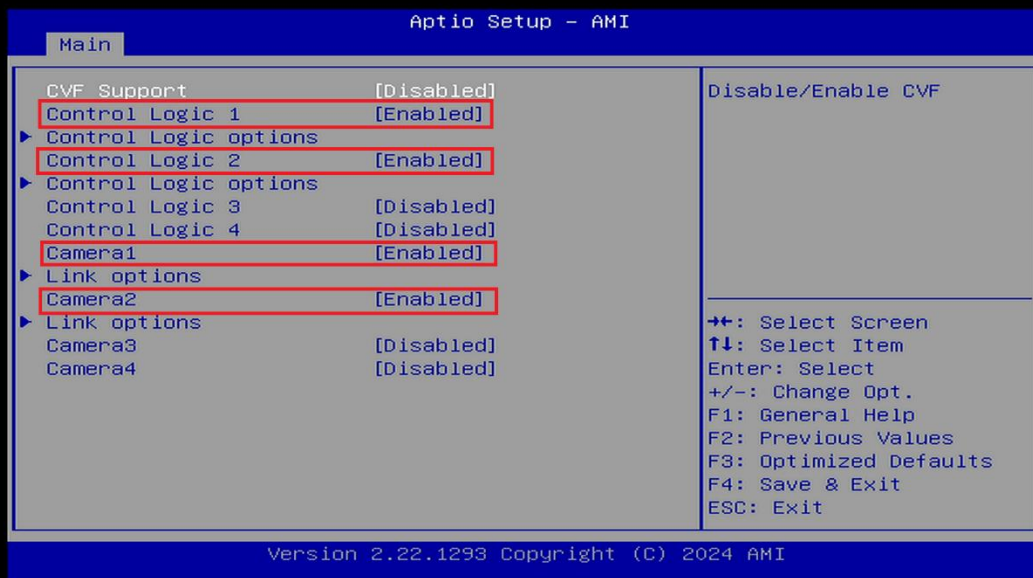
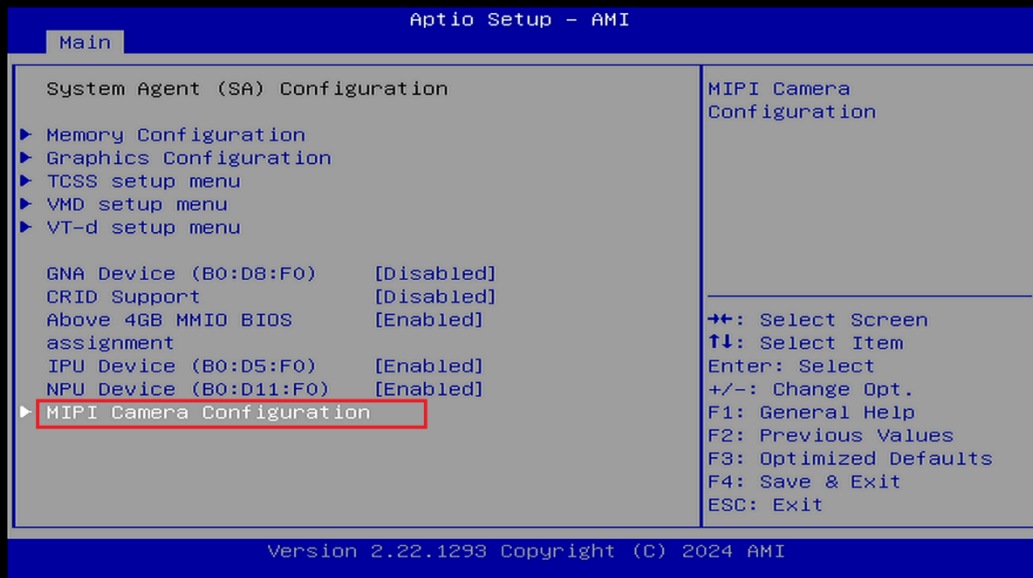
MIPI Camera x 2 (step 1)- (Meteor Lake)



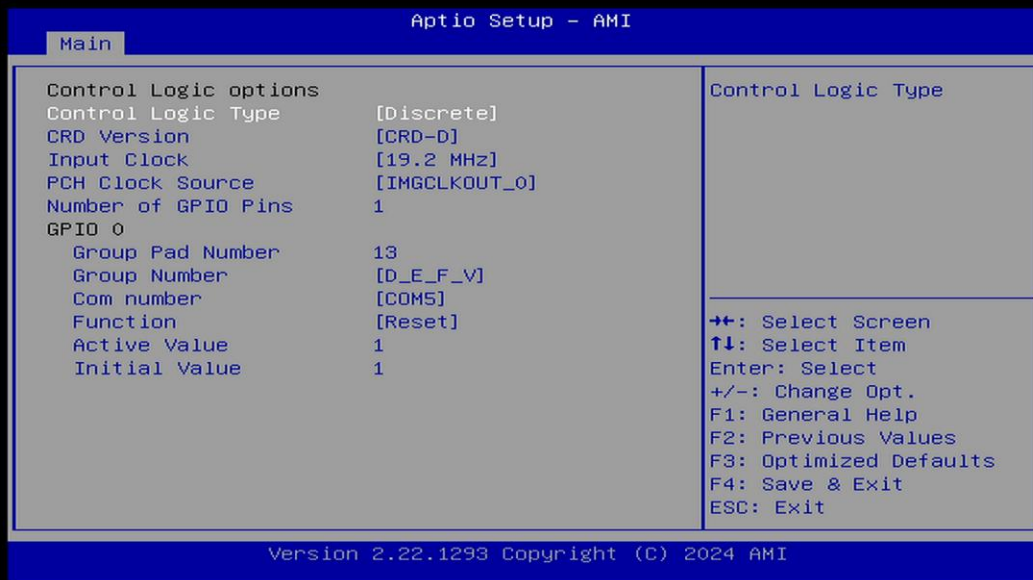
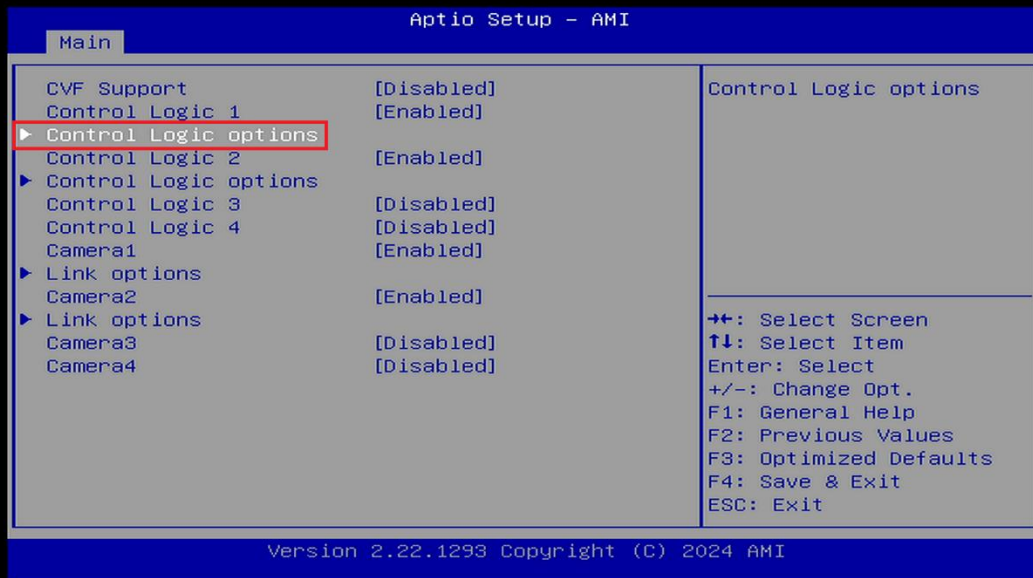
MIPI Camera x 2 (step 2)- (Meteor Lake)



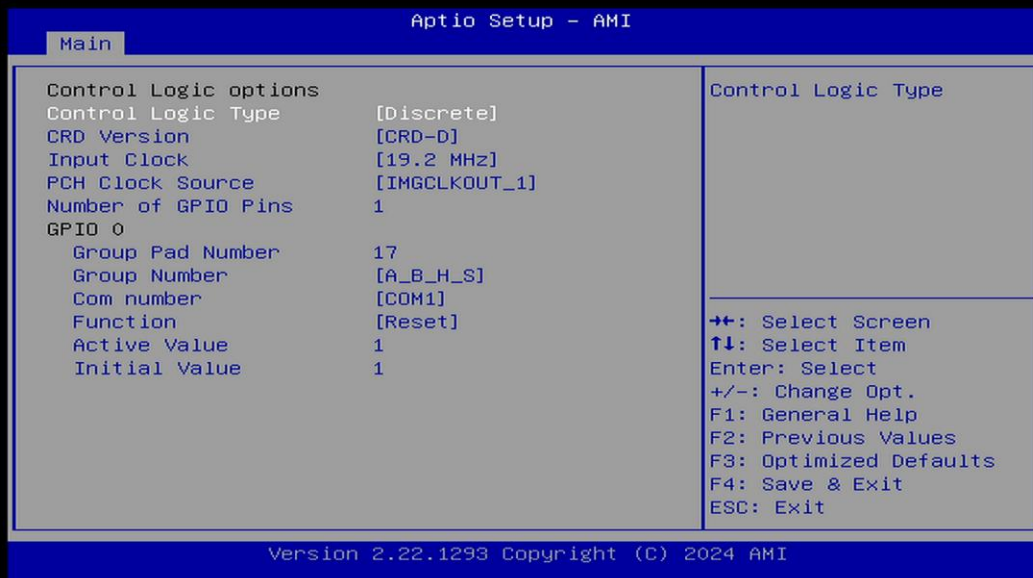
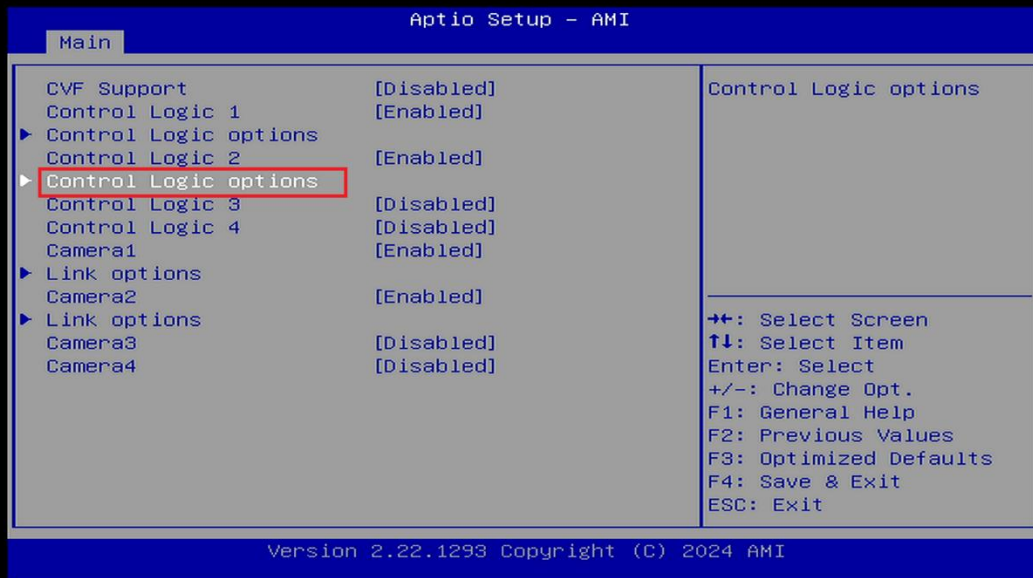
MIPI Camera x 2 (step 3)- (Meteor Lake)



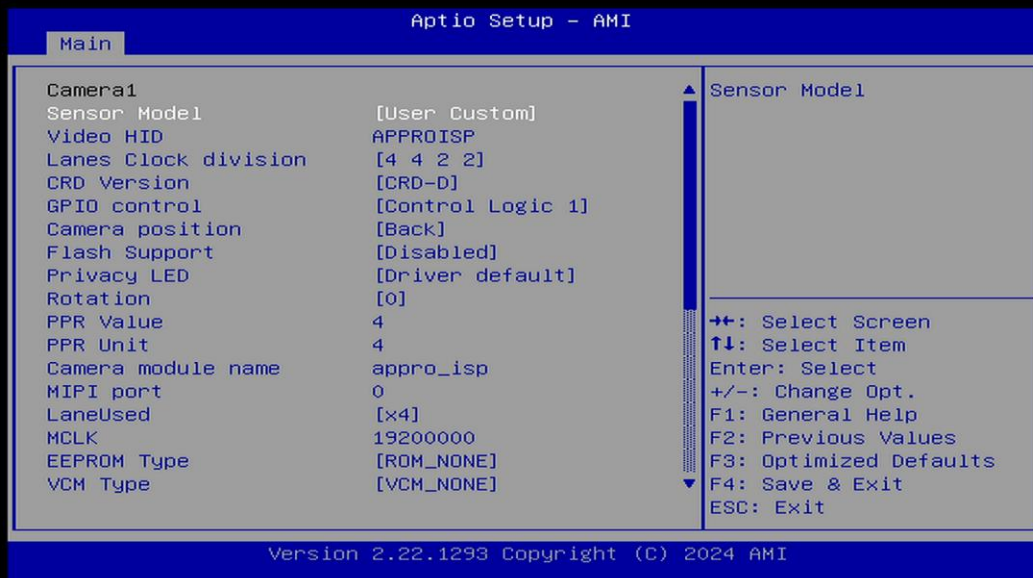
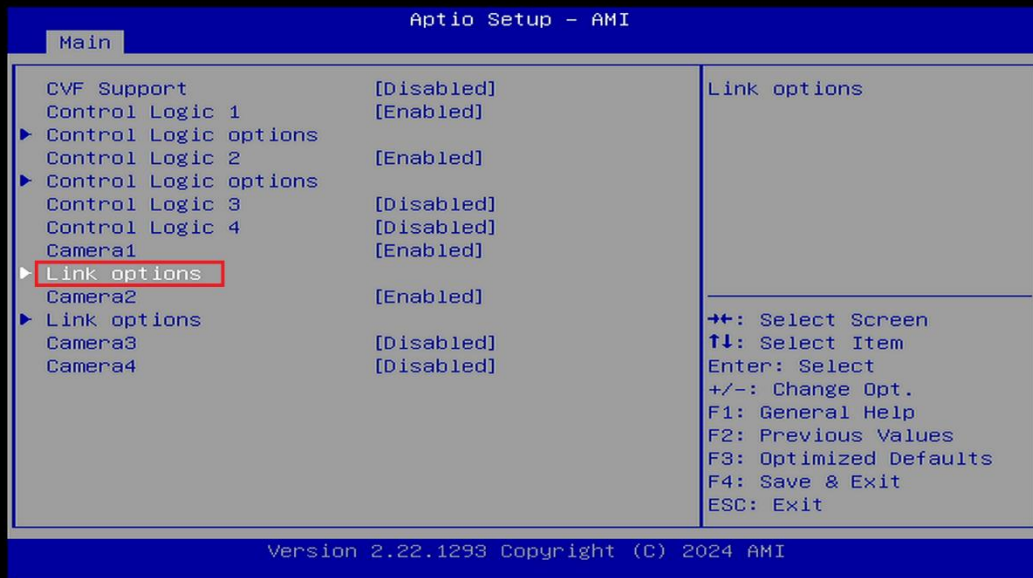
MIPI Camera x 2 (step 4)- (Meteor Lake)



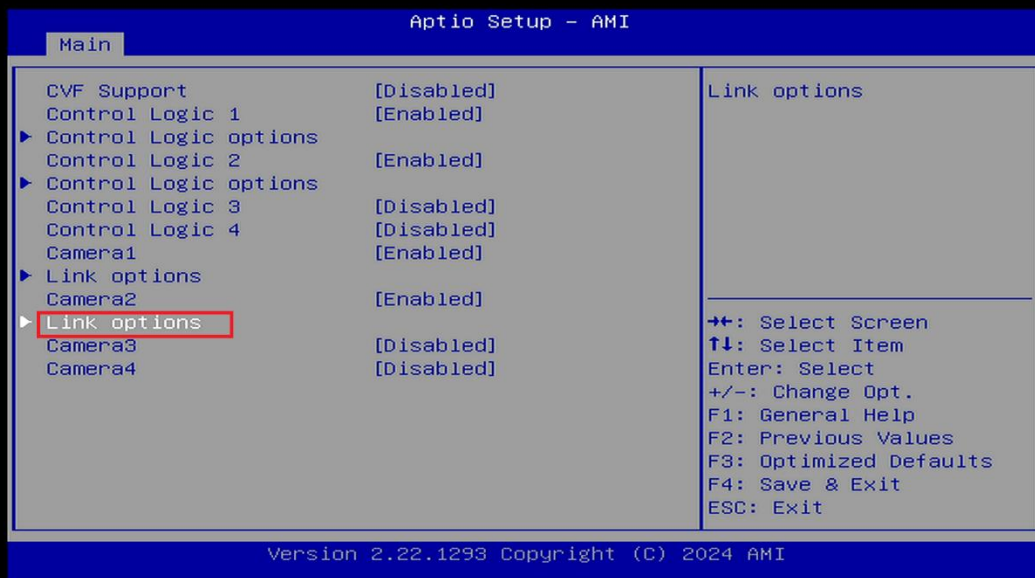
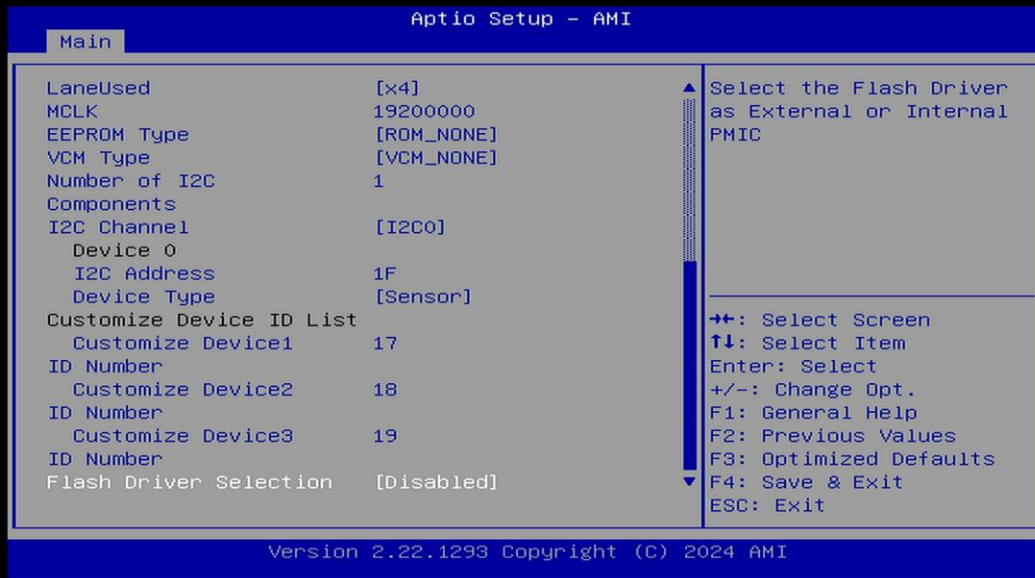
MIPI Camera x 2 (step 5)- (Meteor Lake)



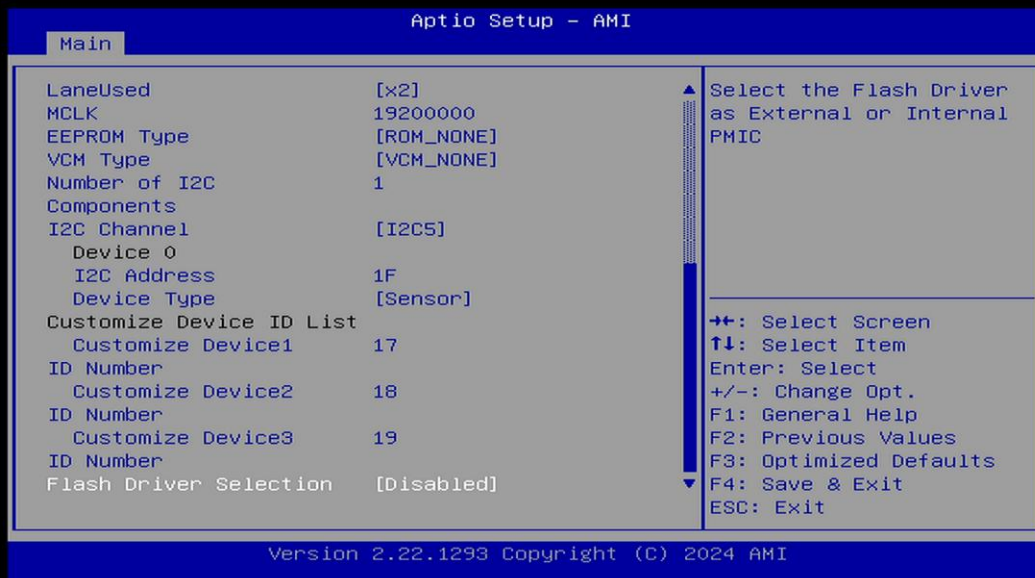
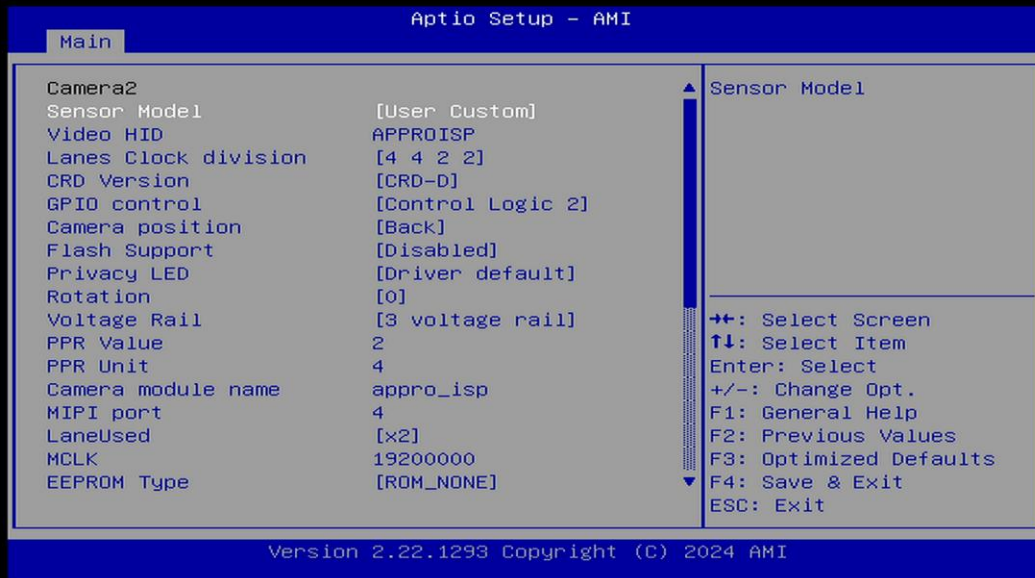
MIPI Camera x 2 (step 6)- (Meteor Lake)



MIPI Camera x 2 (step 7)- (Meteor Lake)



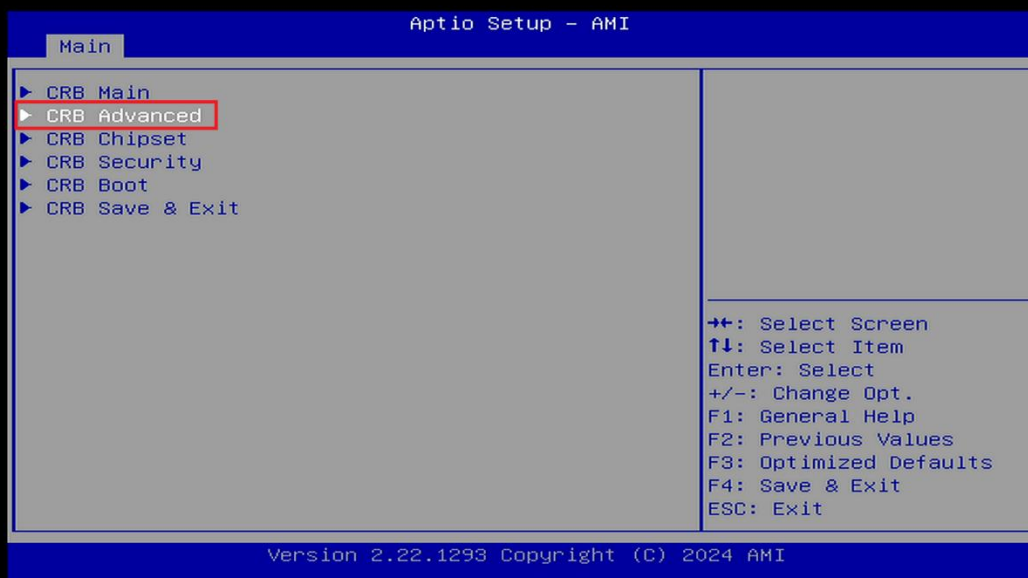
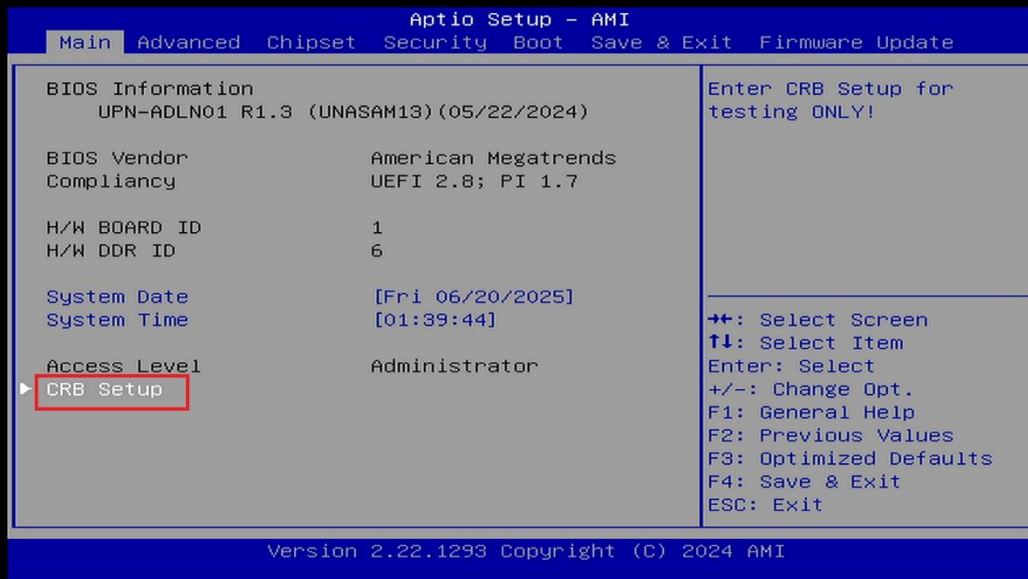
MIPI Camera x 2 (step 8)- (Meteor Lake)



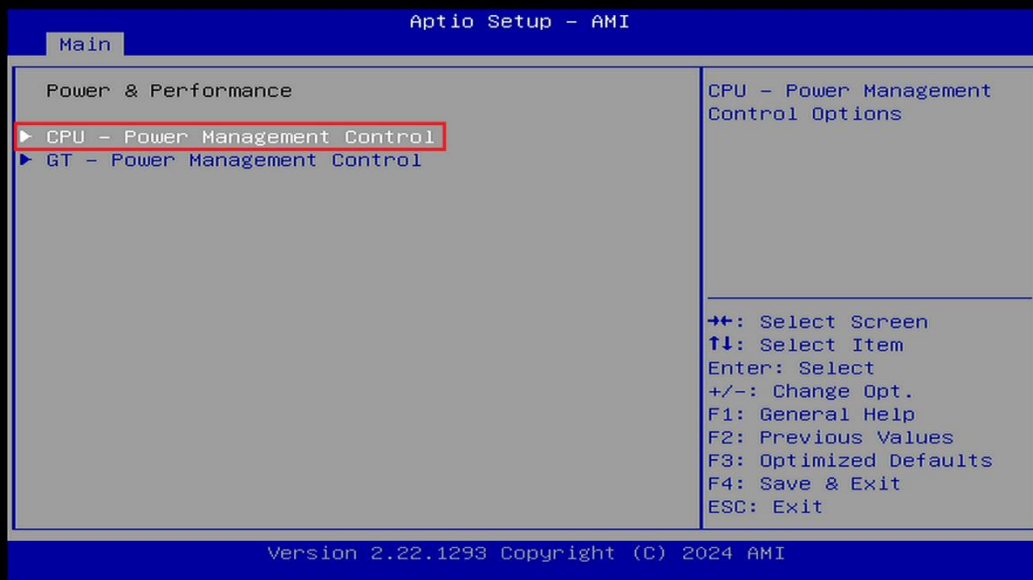
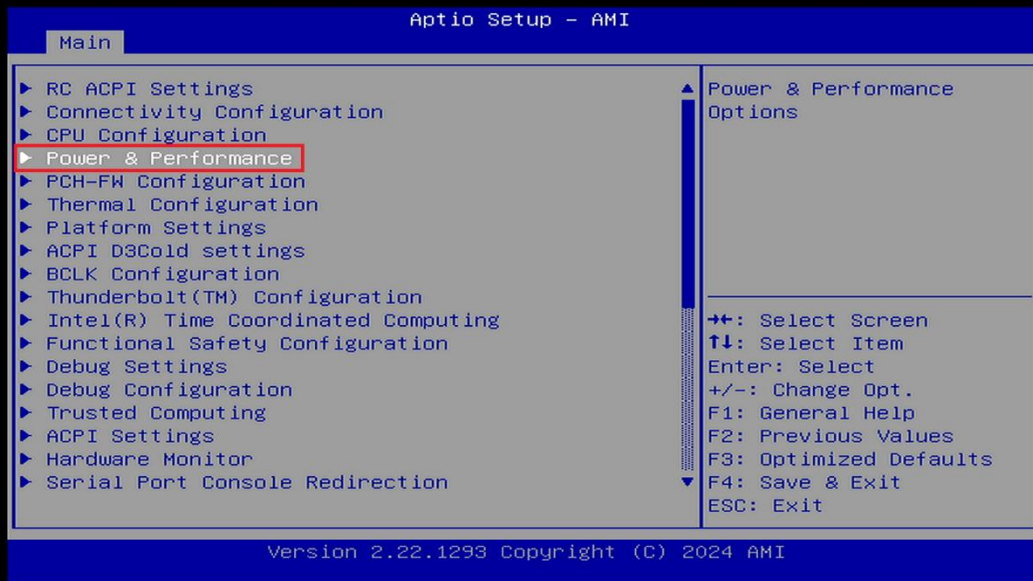
BIOS Setup (Alder Lake)

MISC. Setting

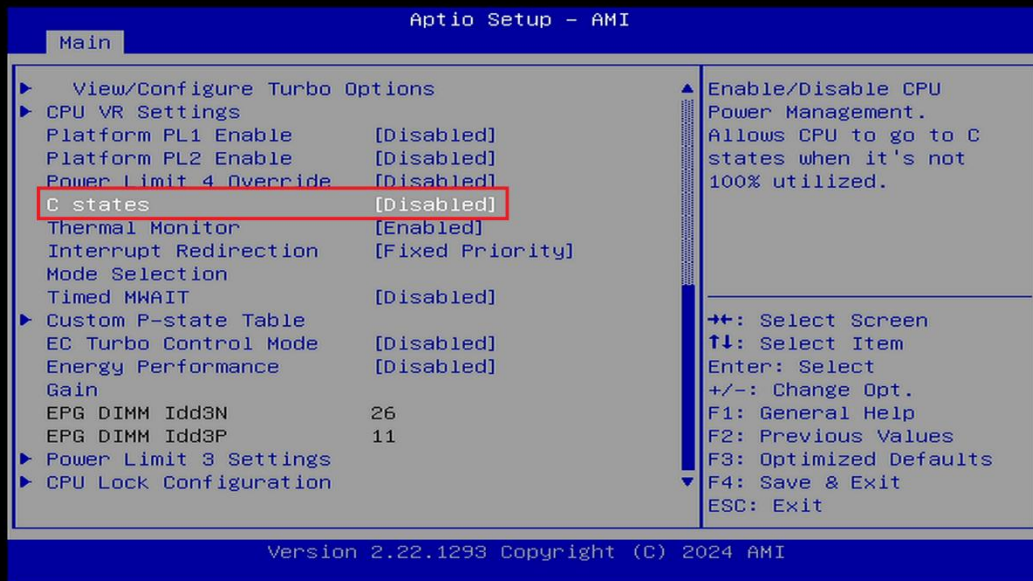
■ Disable C-states (step 1) - (Alder Lake)



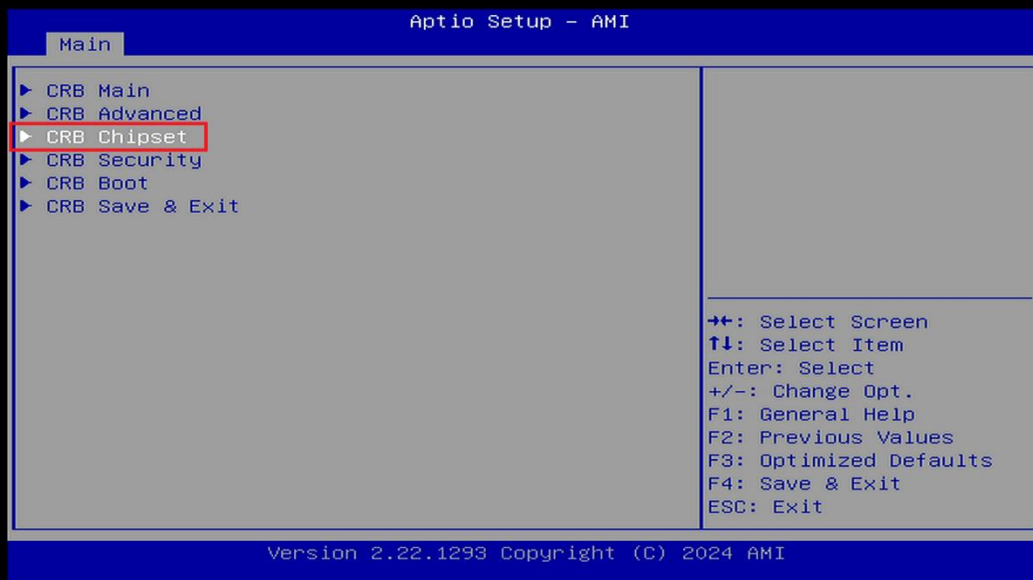
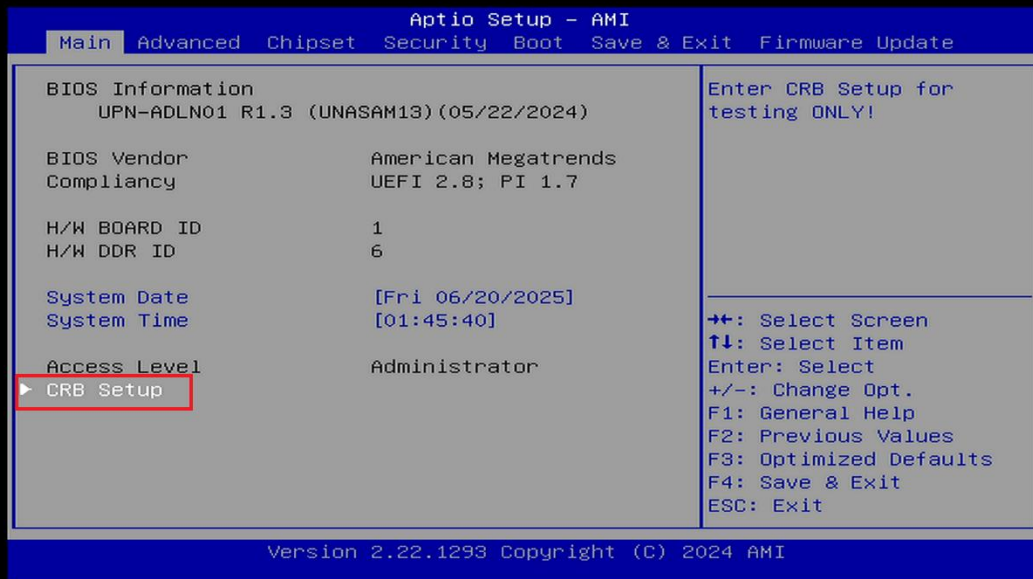
■ Disable C-states (step 2) - (Alder Lake)



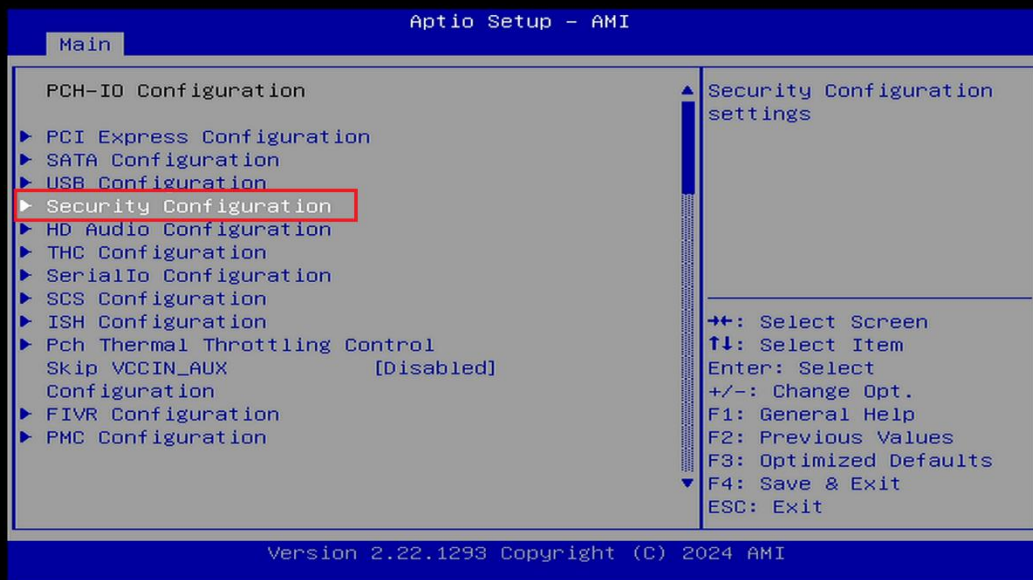
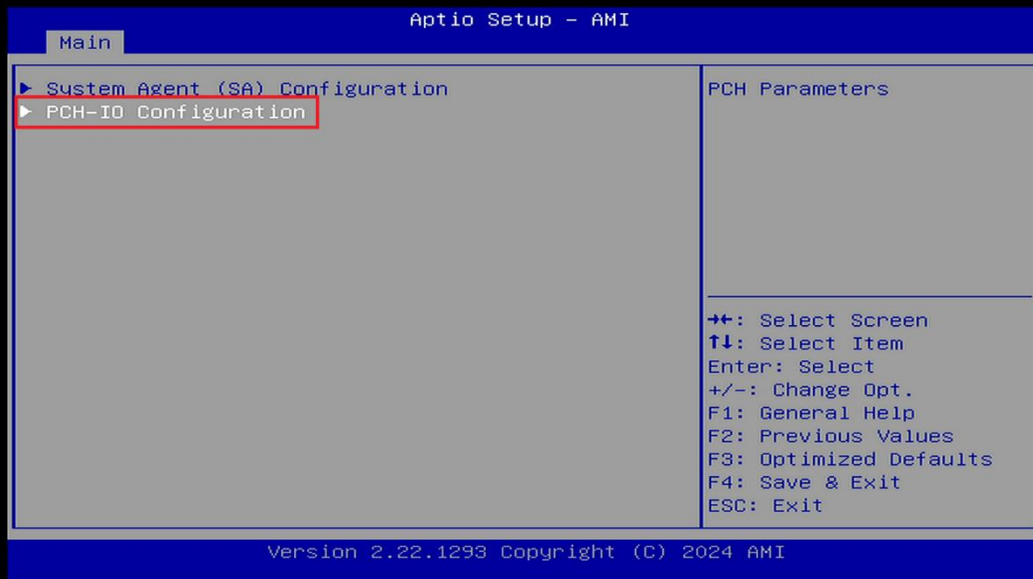
■ Disable C-states (step 3) - (Alder Lake)



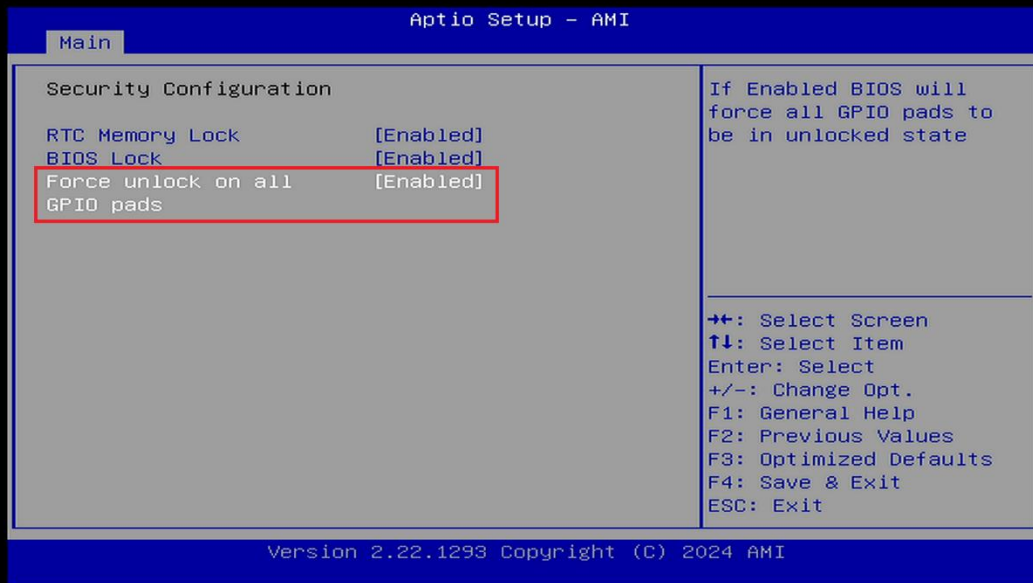
■ Force unlock on all GPIO pads (step 1) - (Alder Lake)



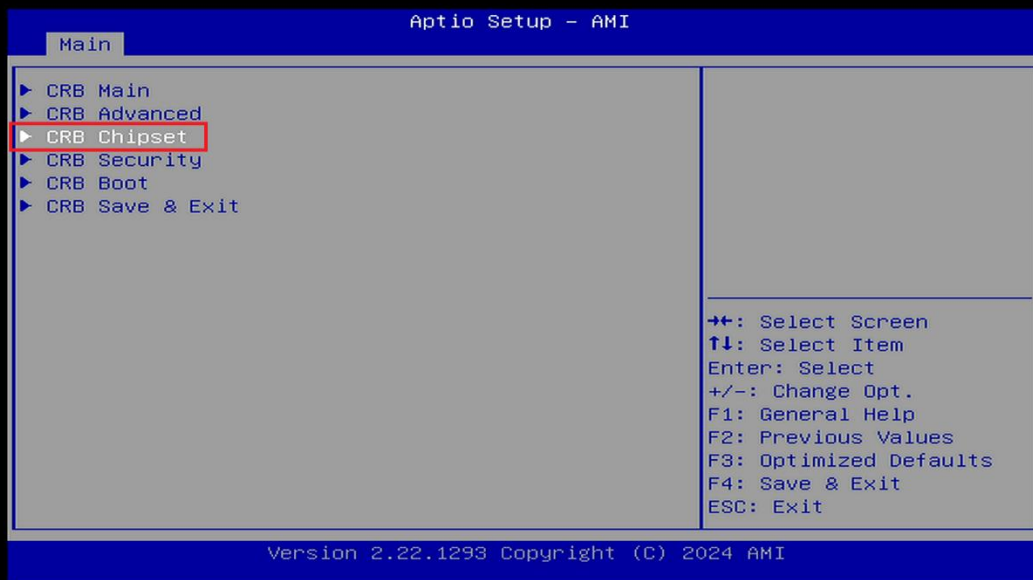
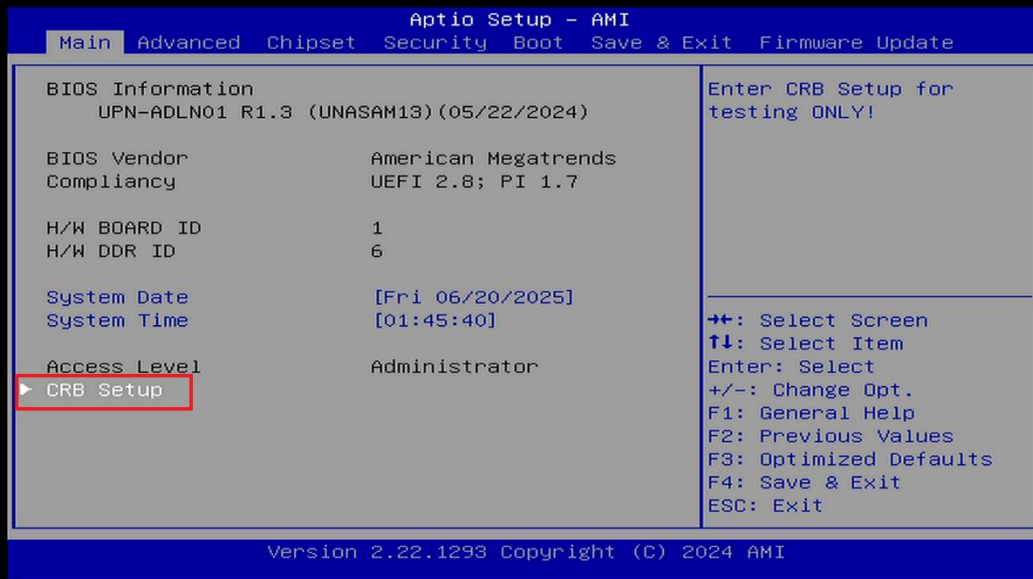
■ Force unlock on all GPIO pads (step 2) - (Alder Lake)



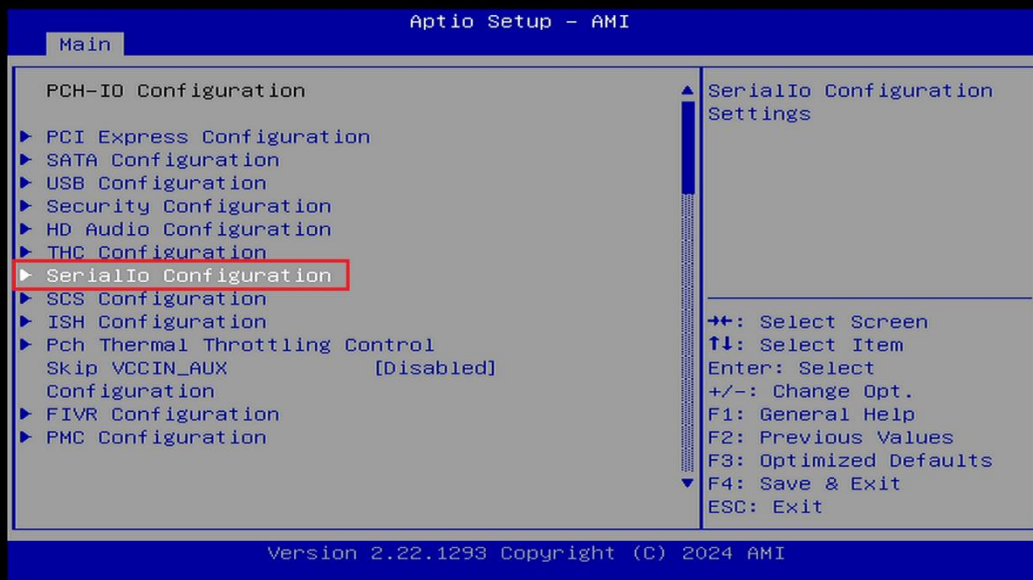
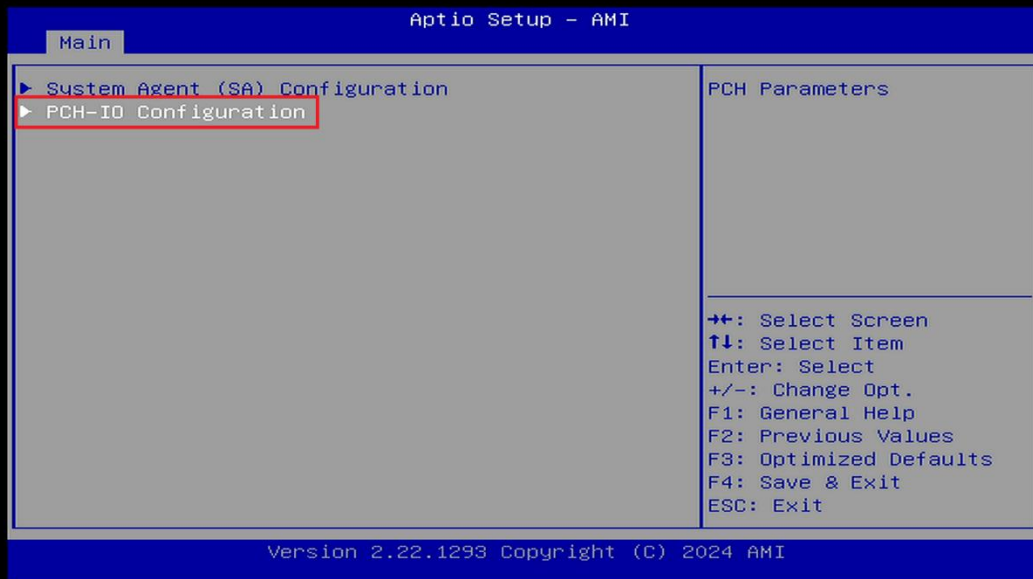
■ Force unlock on all GPIO pads (step 3) - (Alder Lake)



■ I2C Enable (step 1) - (Alder Lake)

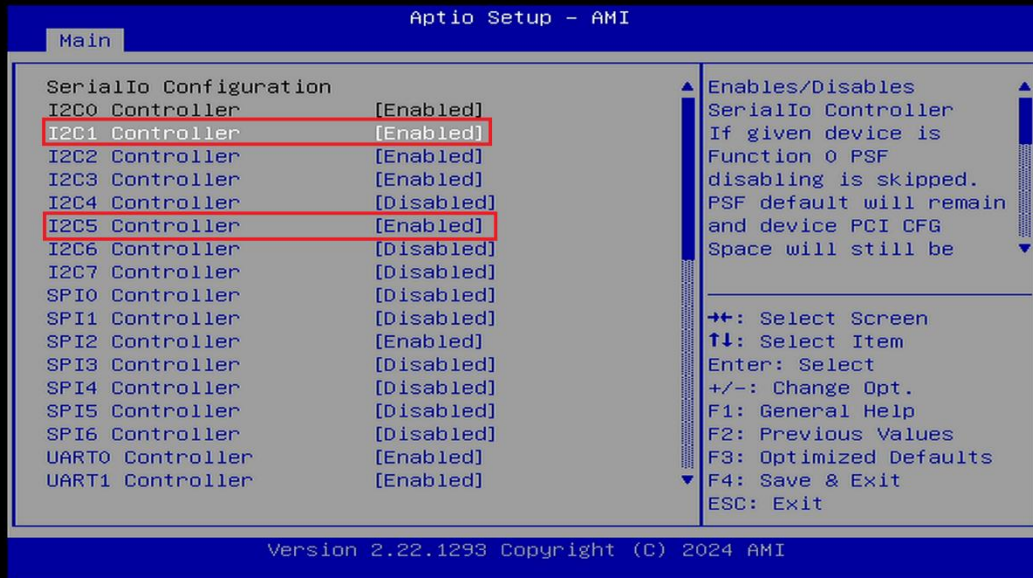


■ I2C Enable (step 2) - (Alder Lake)

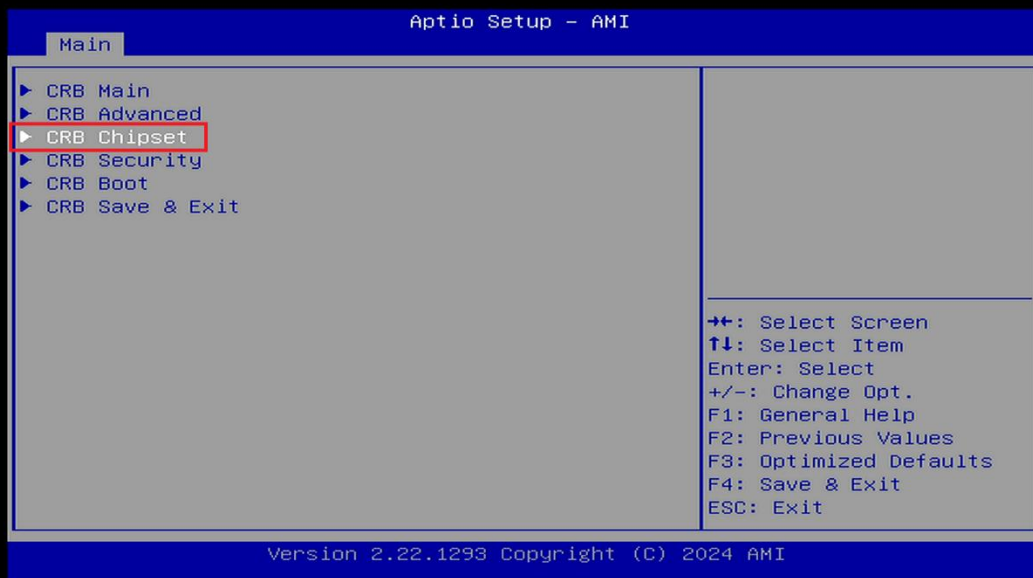
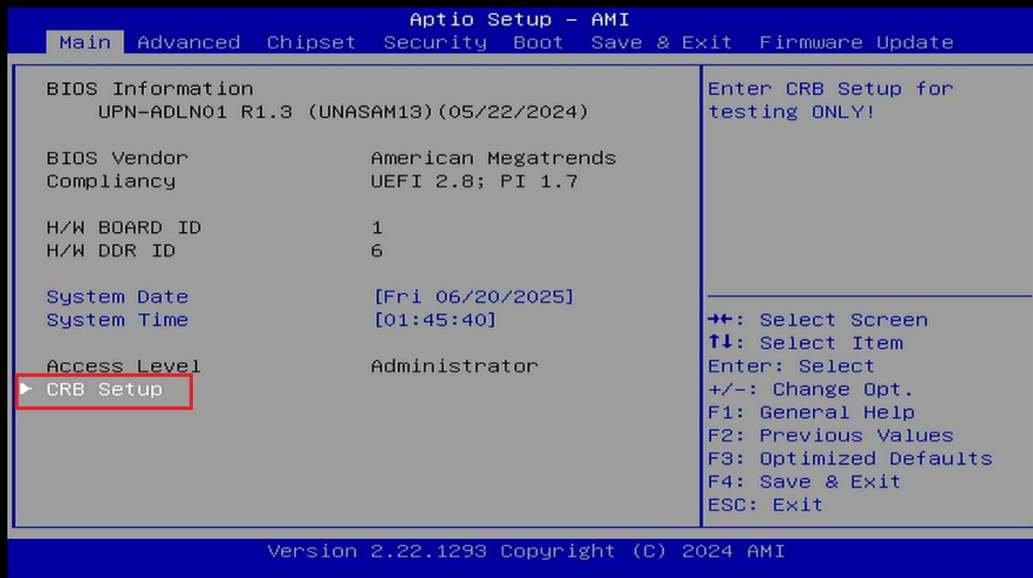


■ I2C Enable (step 3) - (Alder Lake)

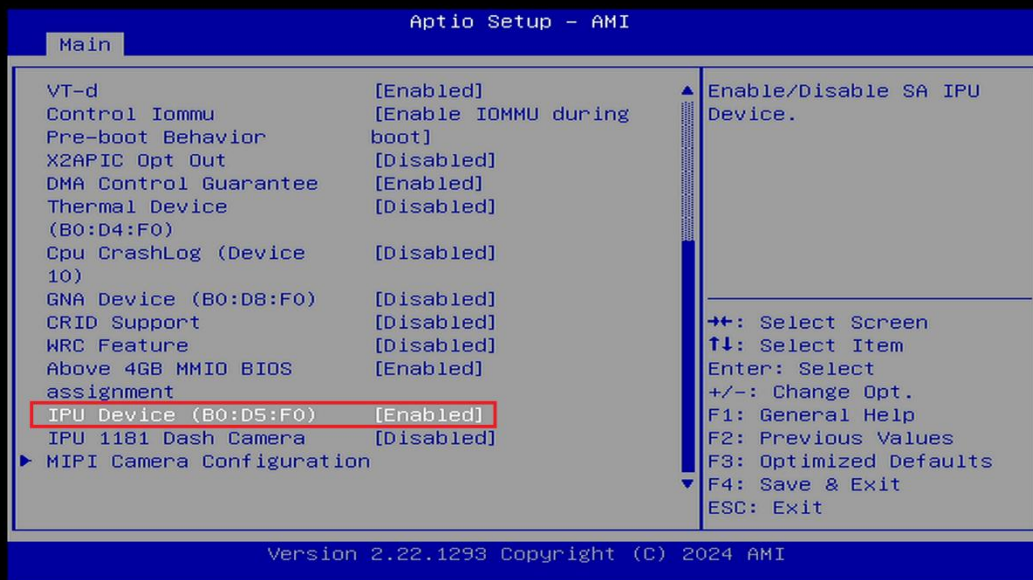
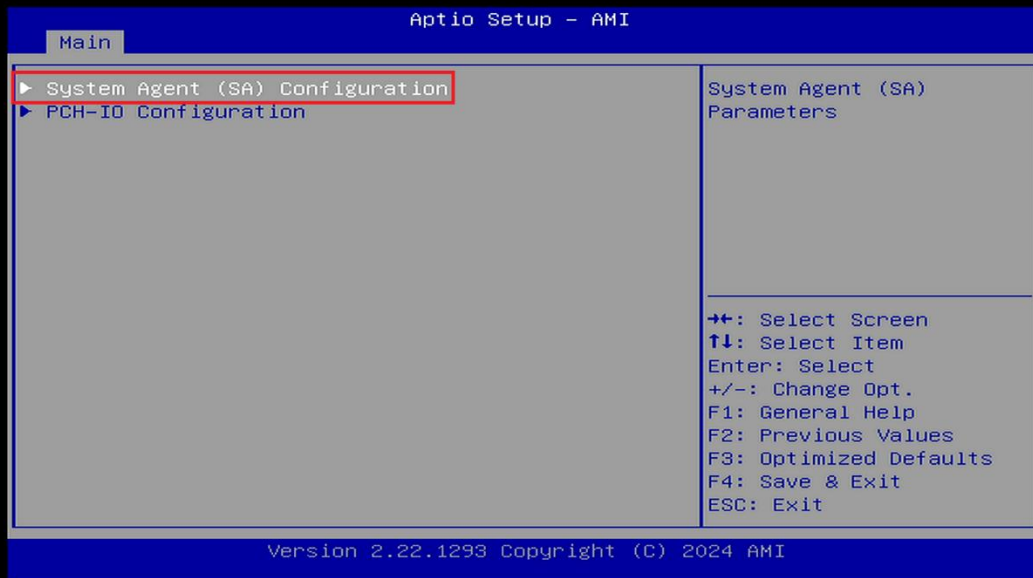
I2C1 and I2C5 must enable.



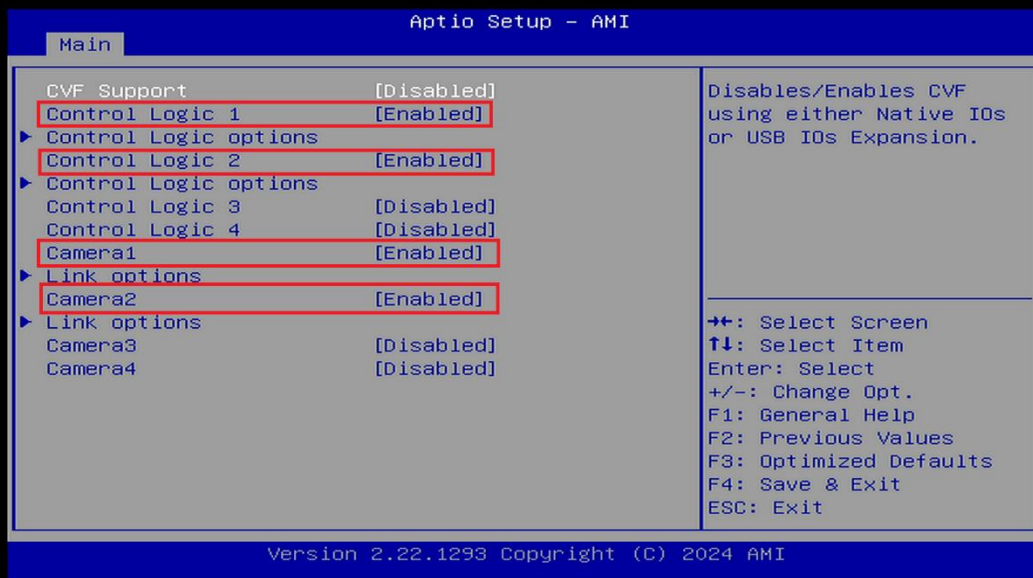
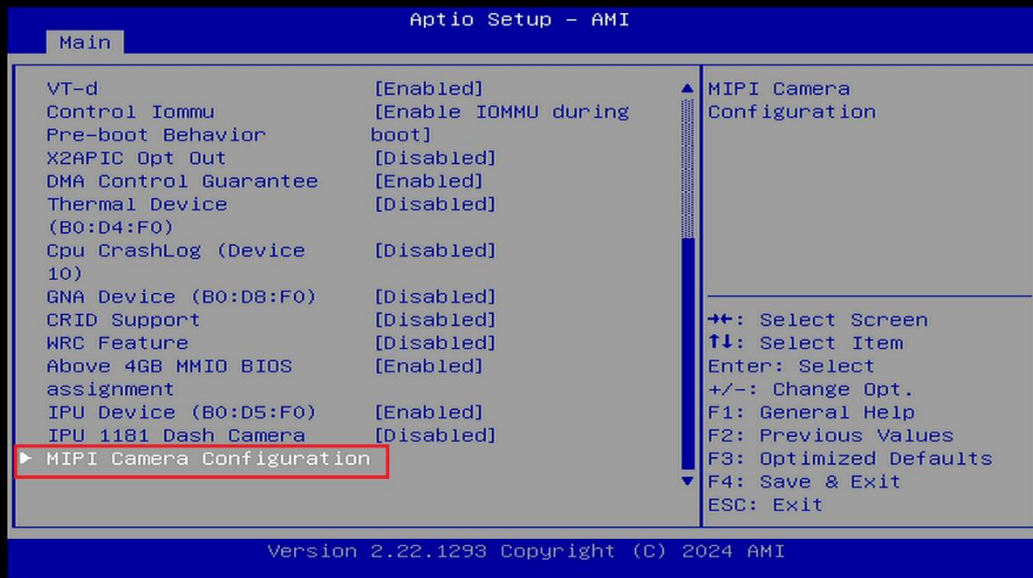
GMSL Camera x 8 (step 1)- (Alder Lake)



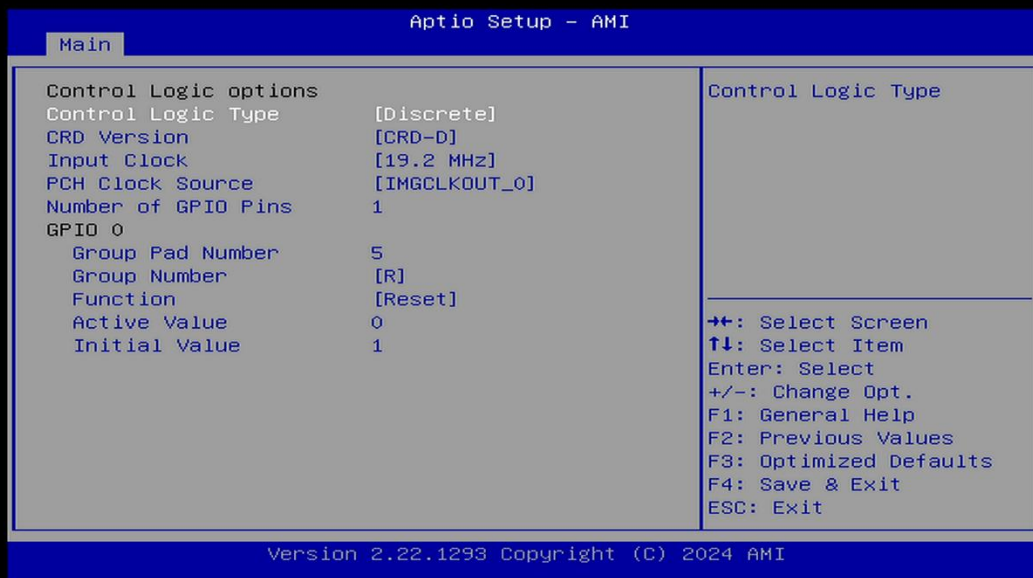
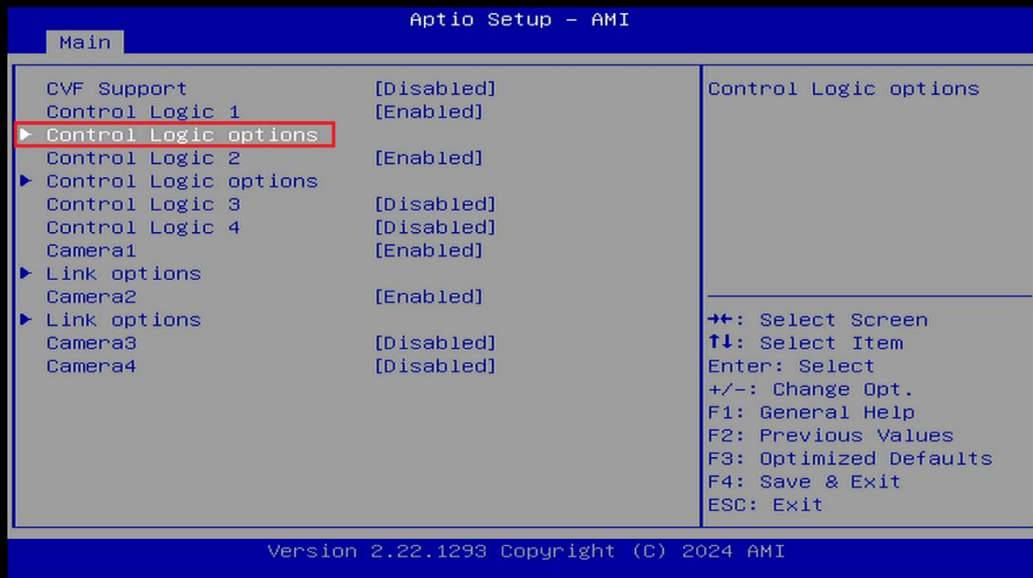
GMSL Camera x 8 (step 2)- (Alder Lake)



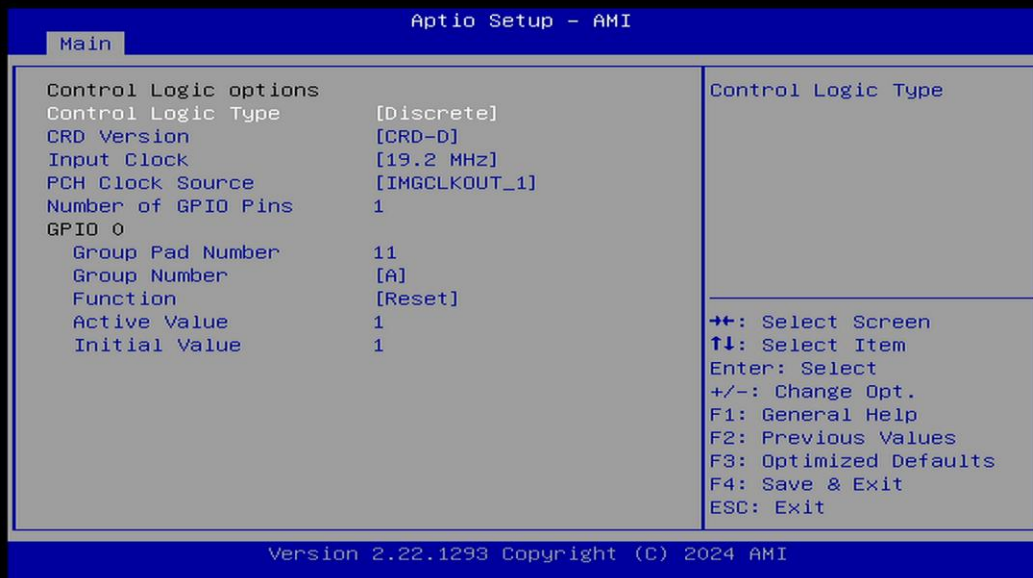
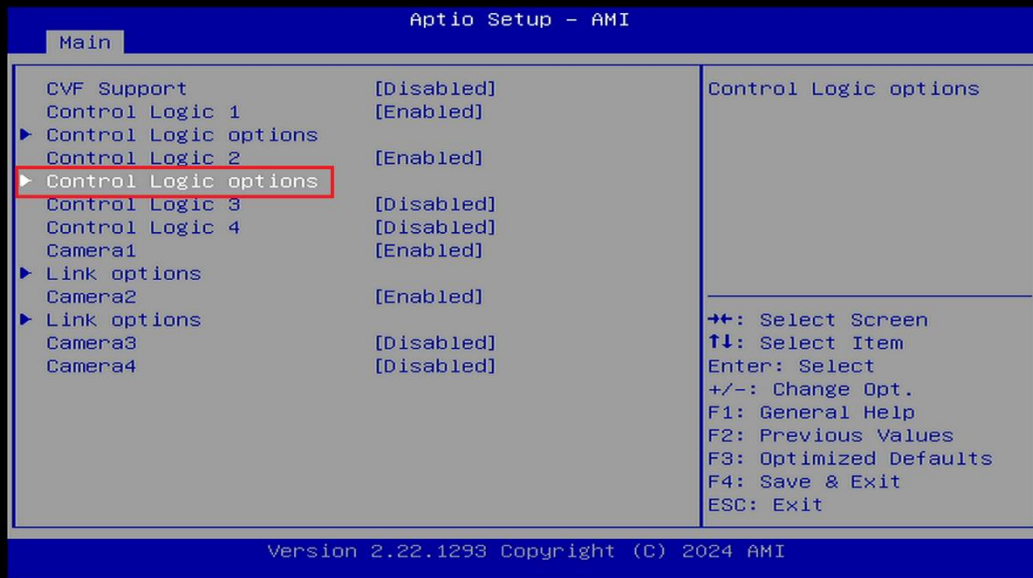
GMSL Camera x 8 (step 3)- (Alder Lake)



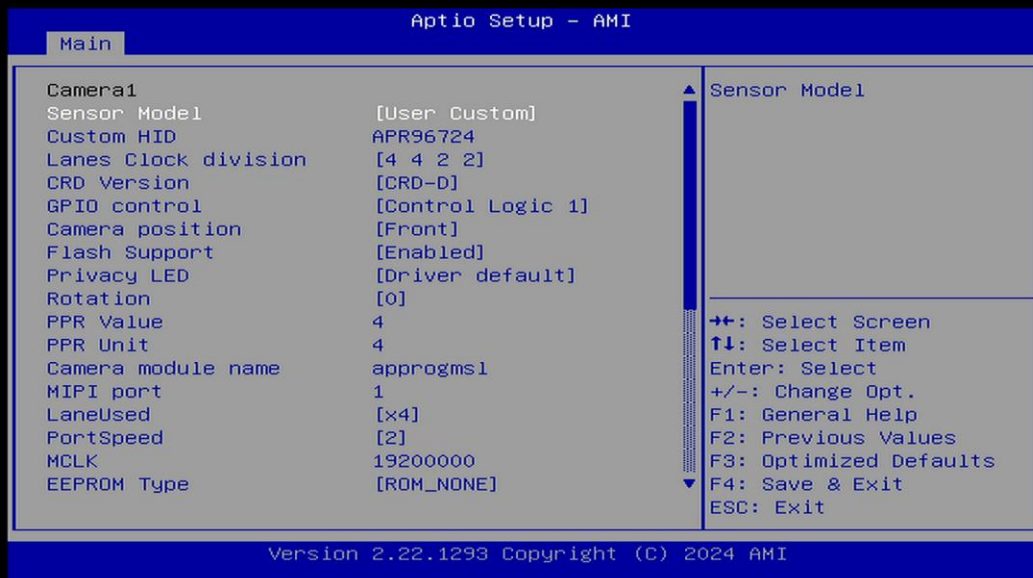
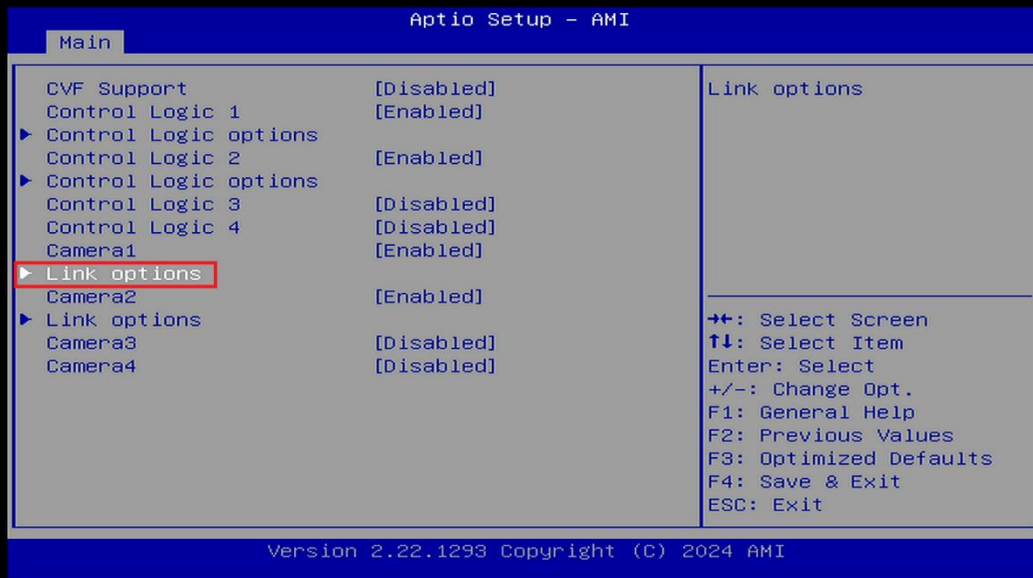
GMSL Camera x 8 (step 4)- (Alder Lake)



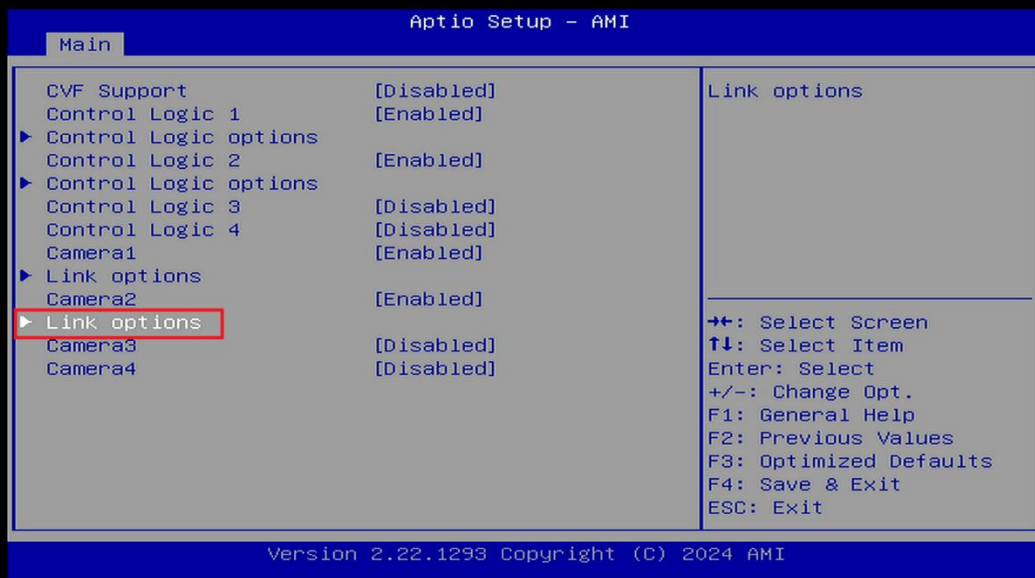
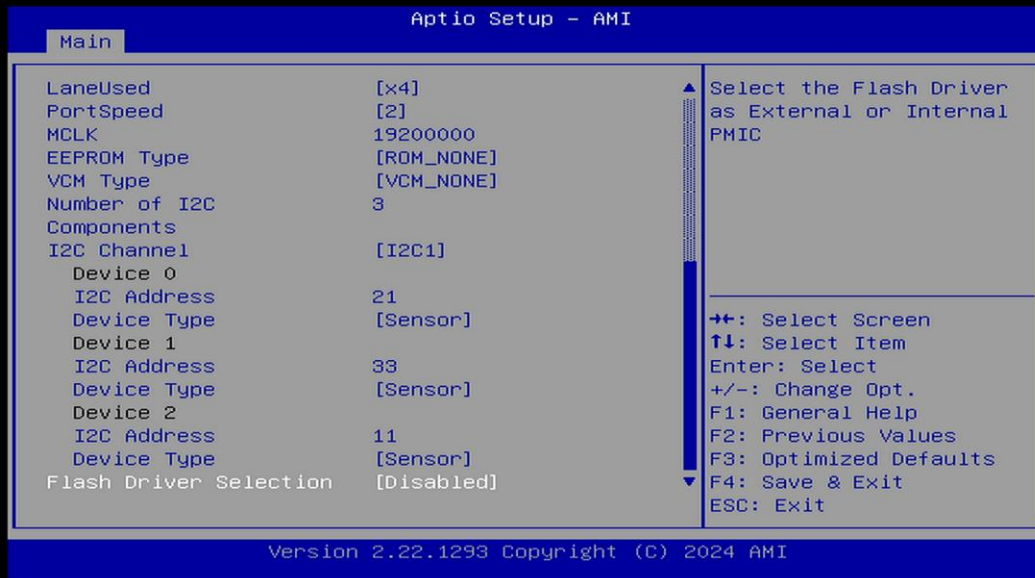
GMSL Camera x 8 (step 5)- (Alder Lake)



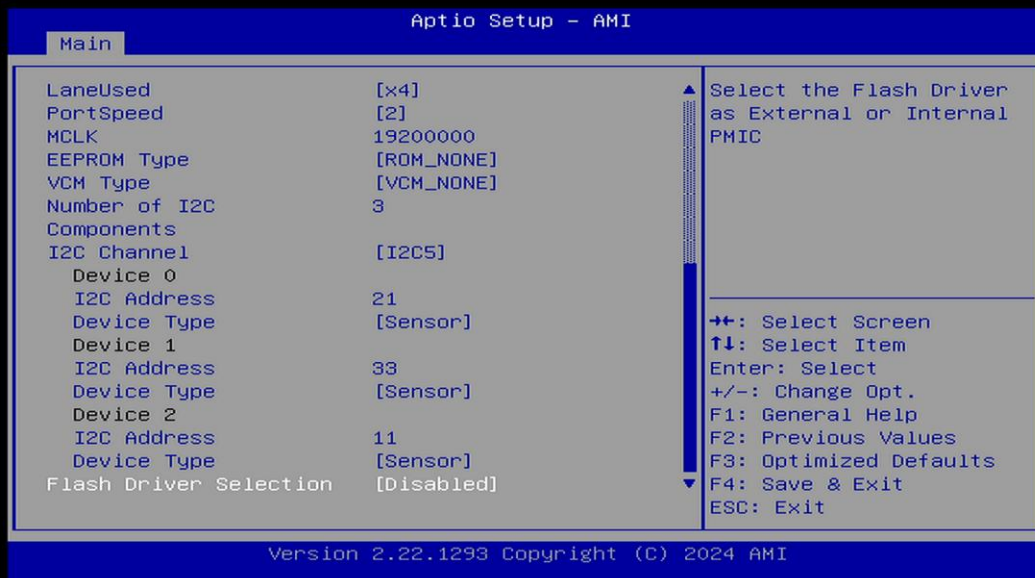
GMSL Camera x 8 (step 6)- (Alder Lake)



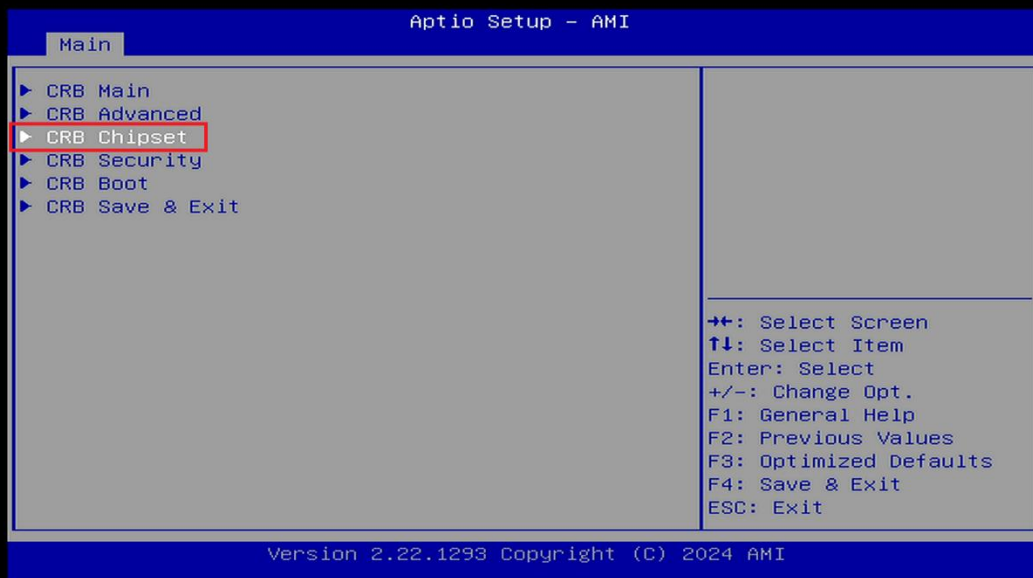
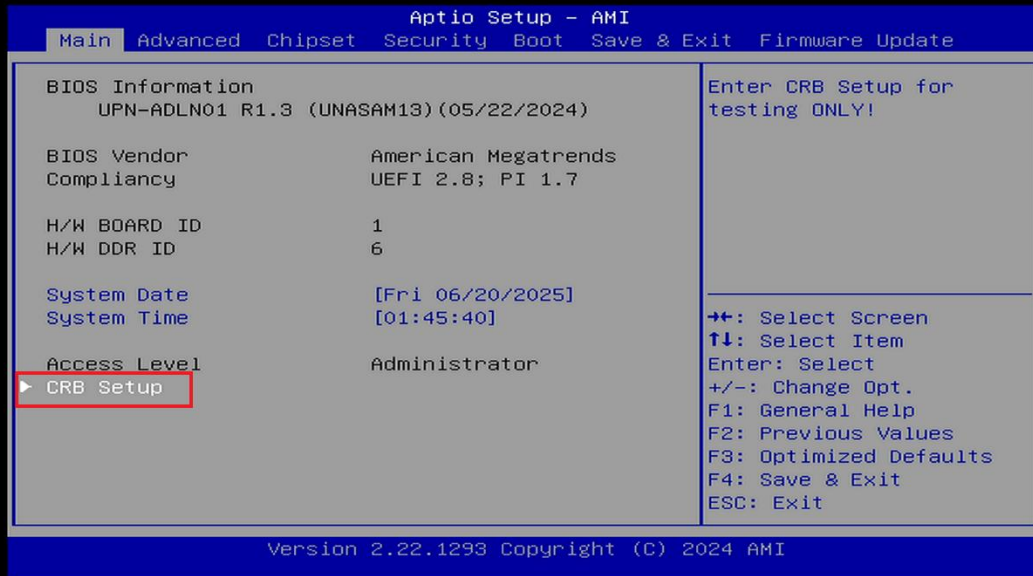
GMSL Camera x 8 (step 7)- (Alder Lake)



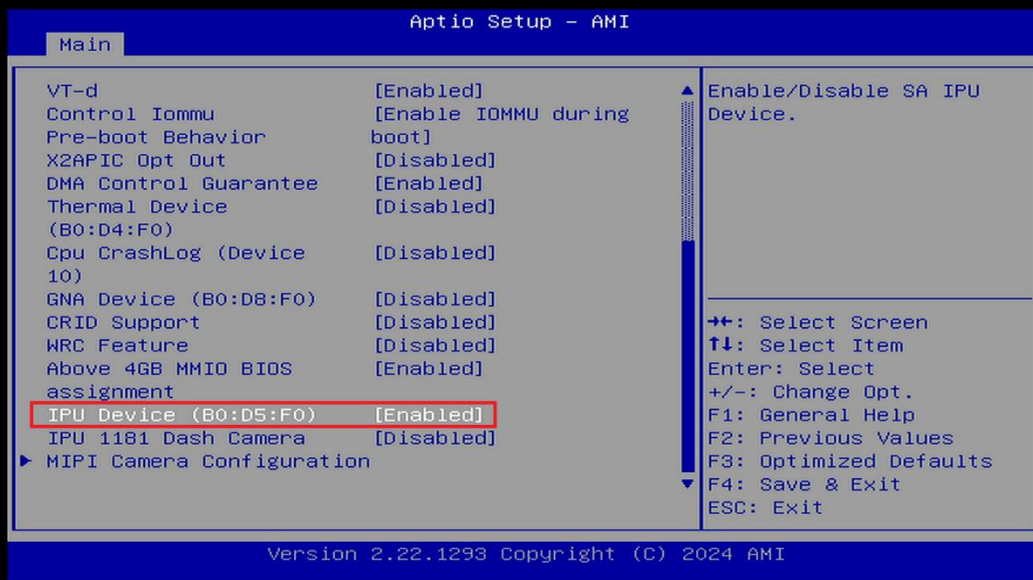
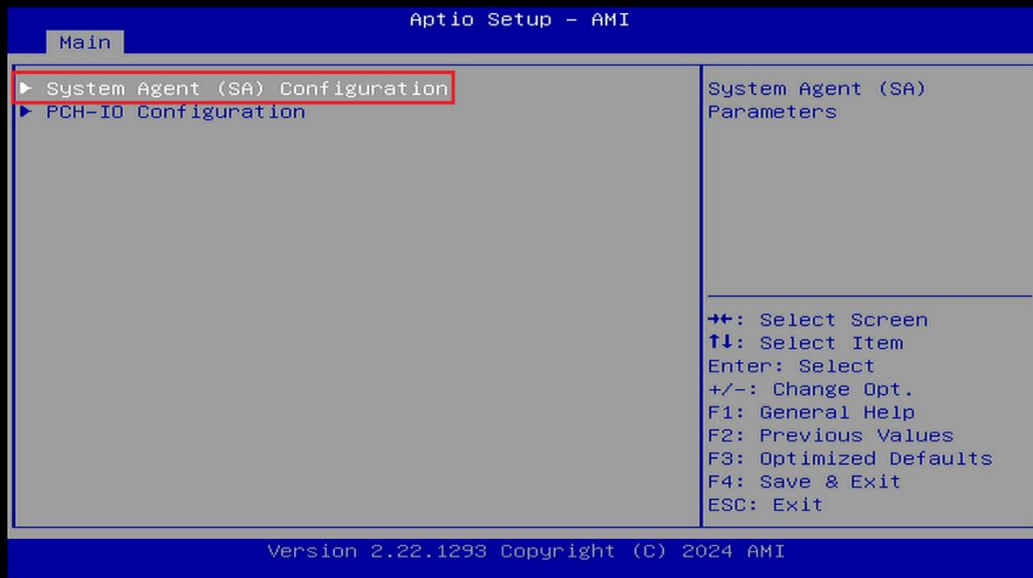
GMSL Camera x 8 (step 8)- (Alder Lake)



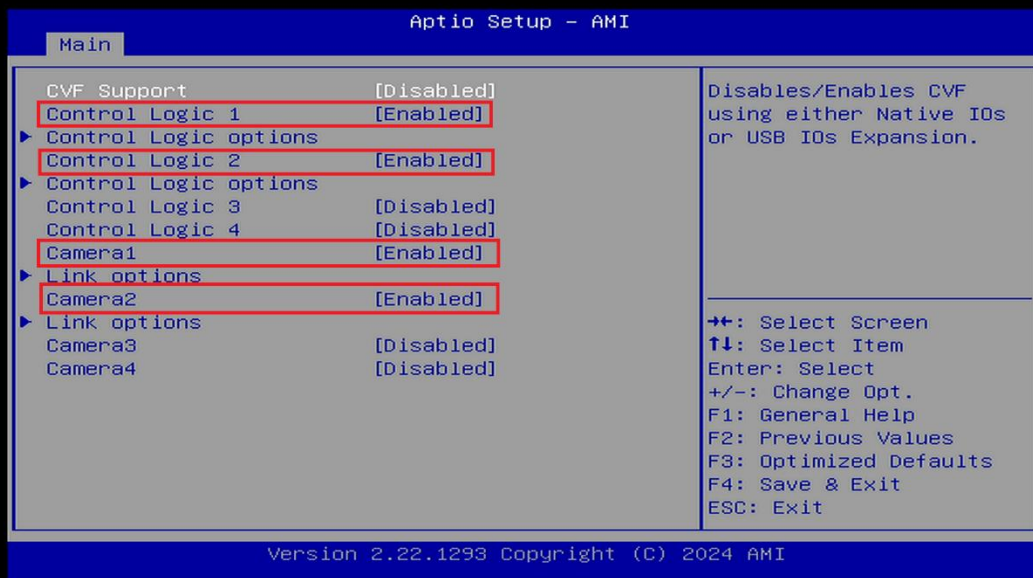
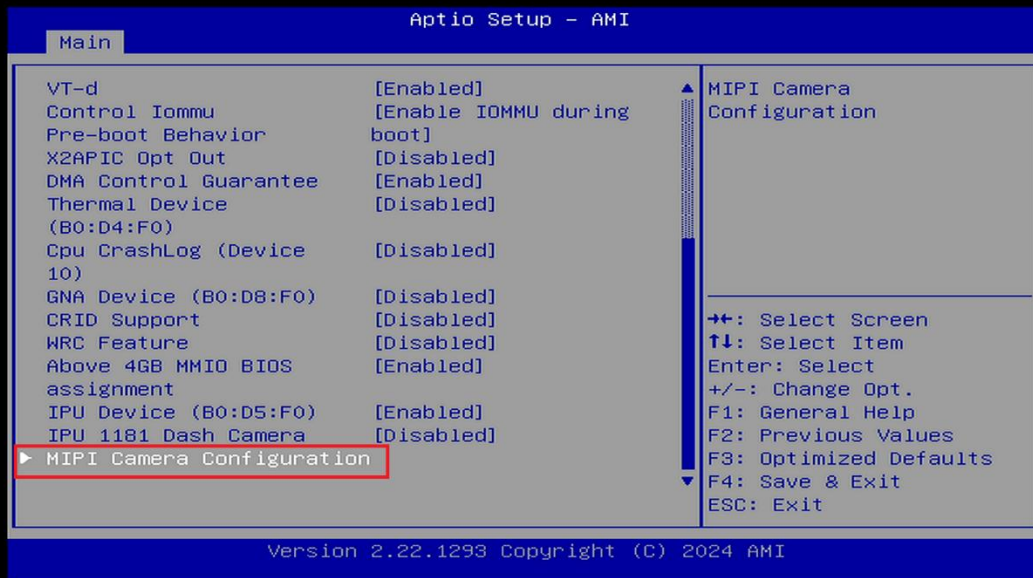
MIPI Camera x 2 (step 1)- (Alder Lake)



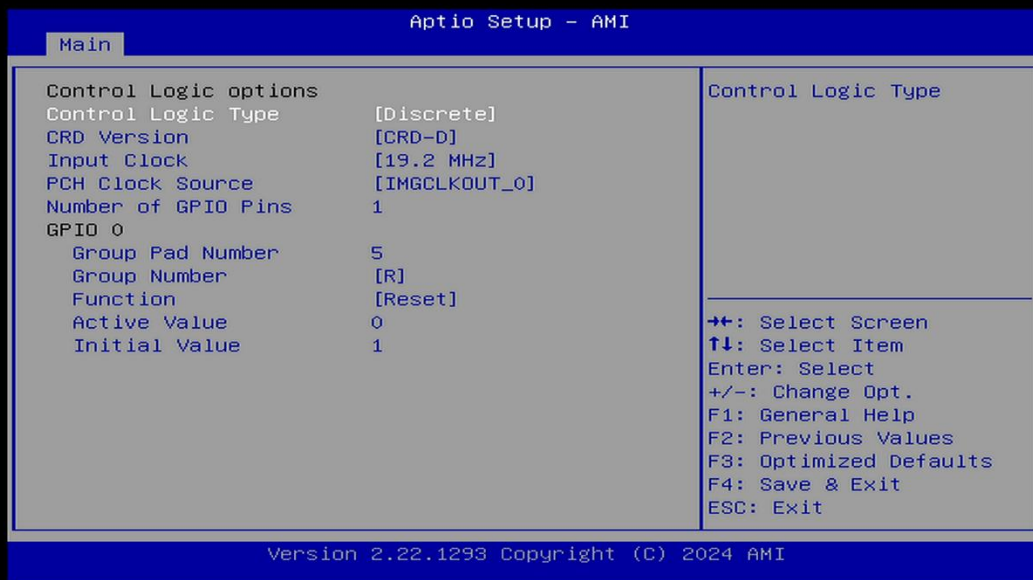
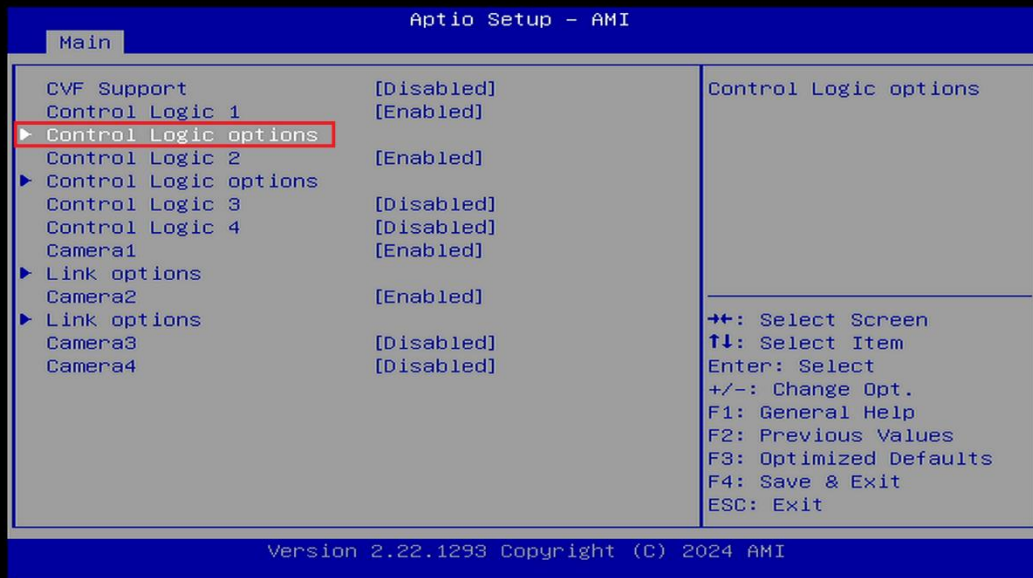
MIPI Camera x 2 (step 2)- (Alder Lake)



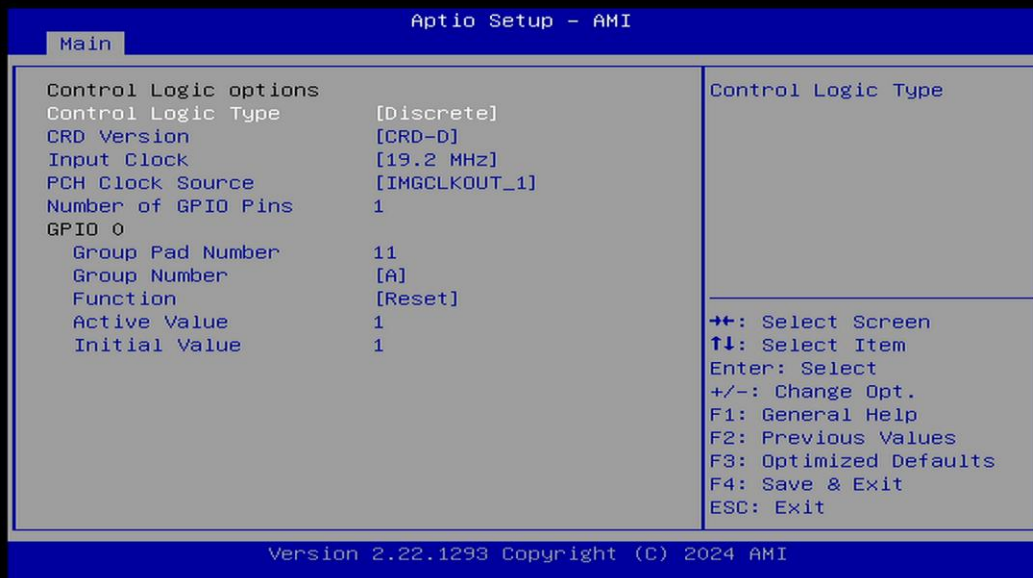
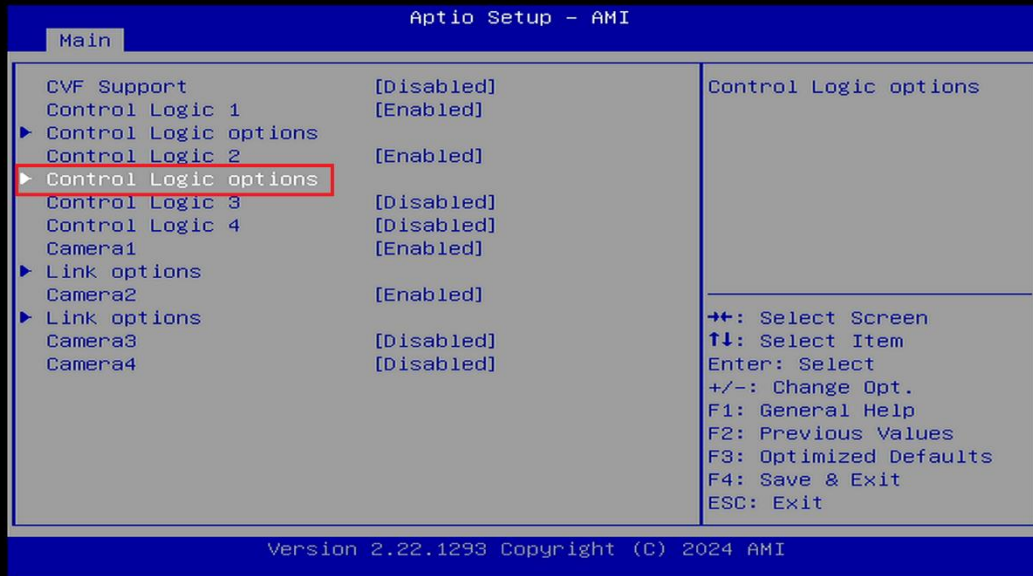
MIPI Camera x 2 (step 3)- (Alder Lake)



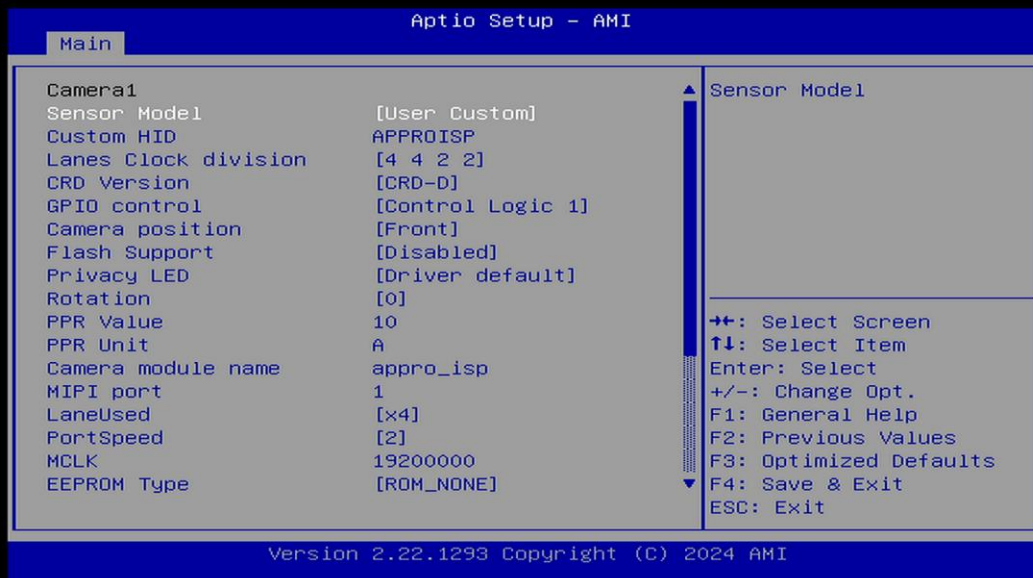
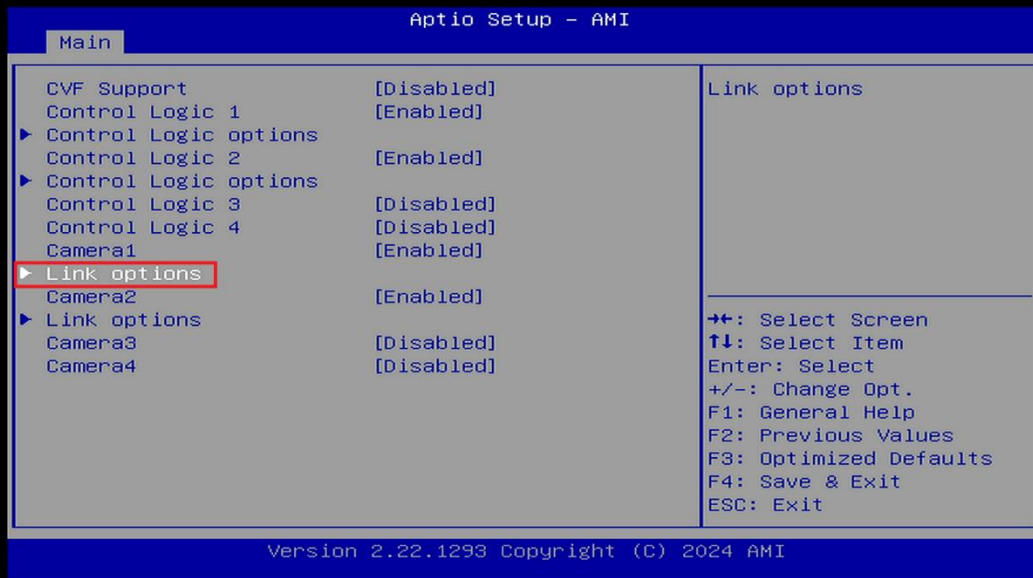
MIPI Camera x 2 (step 4)- (Alder Lake)



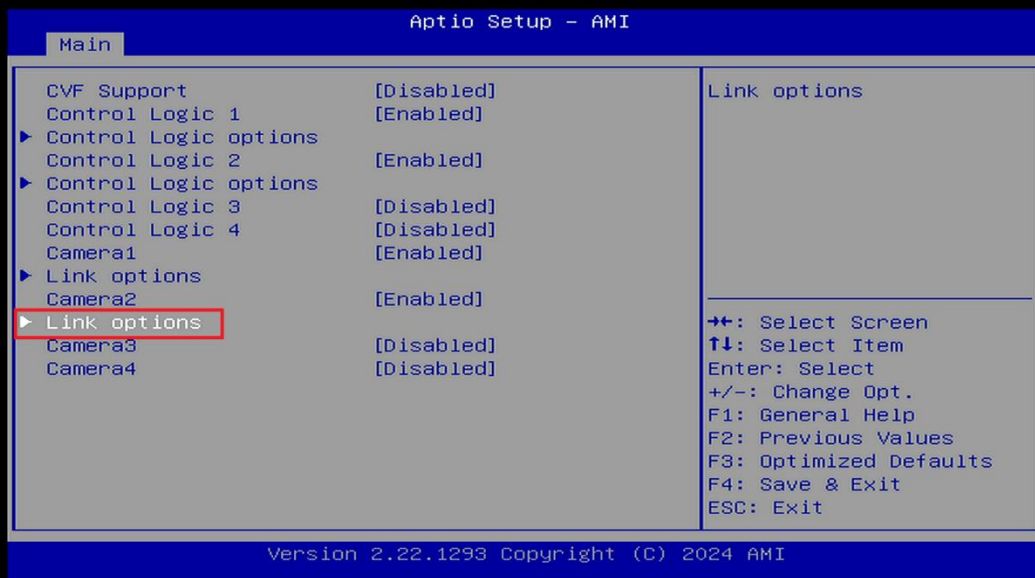
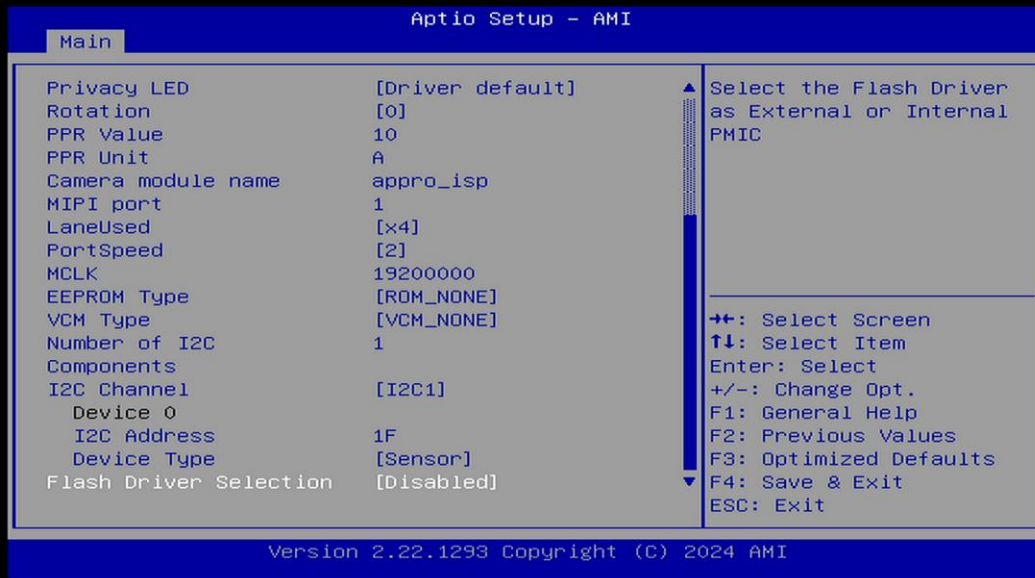
MIPI Camera x 2 (step 5)- (Alder Lake)



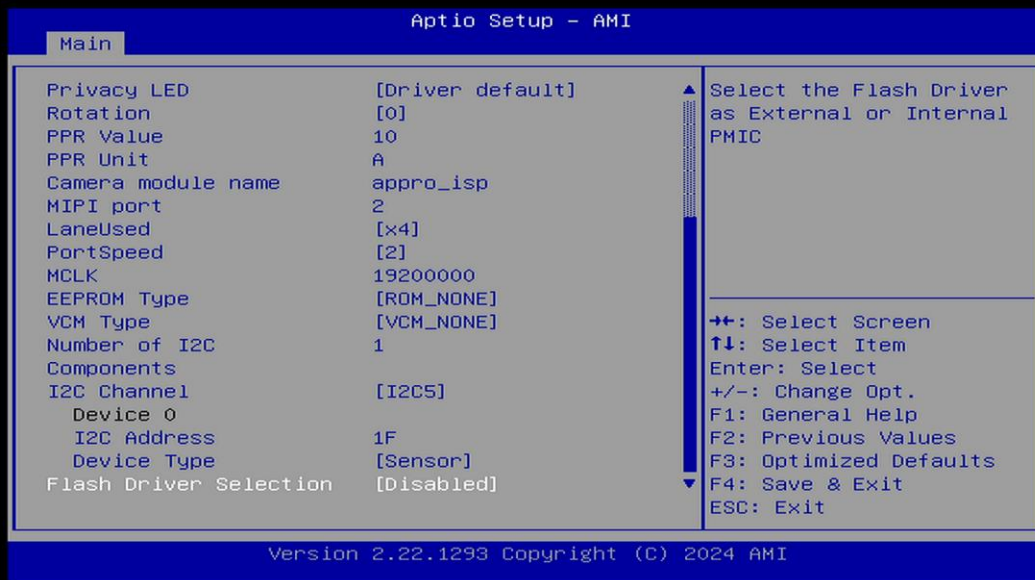
MIPI Camera x 2 (step 6)- (Alder Lake)



MIPI Camera x 2 (step 7)- (Alder Lake)



MIPI Camera x 2 (step 8)- (Alder Lake)



Camera Verification

Camera List Command

If driver is installed properly, camera name can be listed in below command.

■ GMSL x 8

```
$ grep . /sys/class/video4linux/v4l-subdev*/name | grep -E 'APPRO_CAM|approgmsl'  
/sys/class/video4linux/v4l-subdev11/name:approgmsl a  
/sys/class/video4linux/v4l-subdev12/name:approgmsl b  
/sys/class/video4linux/v4l-subdev13/name:approgmsl c  
/sys/class/video4linux/v4l-subdev14/name:approgmsl d  
/sys/class/video4linux/v4l-subdev16/name:approgmsl e  
/sys/class/video4linux/v4l-subdev17/name:approgmsl f  
/sys/class/video4linux/v4l-subdev18/name:approgmsl g  
/sys/class/video4linux/v4l-subdev19/name:approgmsl h
```

■ MIPI x 2

```
$ grep . /sys/class/video4linux/v4l-subdev*/name | grep -E 'APPRO_CAM|approgmsl'  
/sys/class/video4linux/v4l-subdev11/name:APPRO_CAM a  
/sys/class/video4linux/v4l-subdev12/name:APPRO_CAM e
```

Camera Preview Test Command

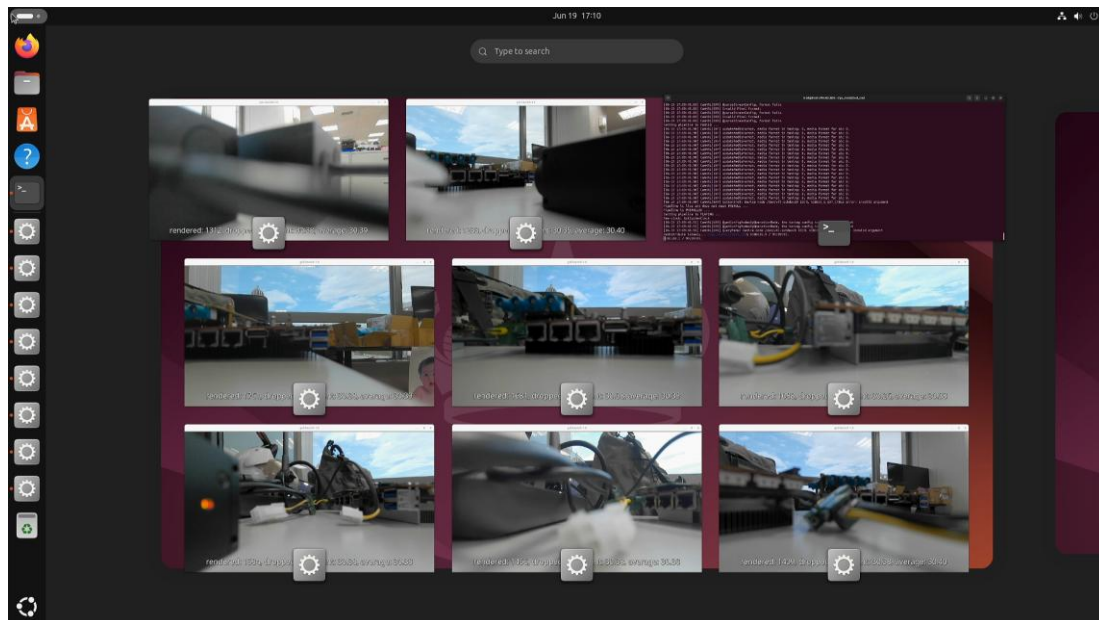
■ GMSL x 8

■ Meteor Lake

```
$ cd release_package/ubuntu/test_cmd/  
$ ./mtl_gmsl_up_abcdefgh.sh
```

■ Alder Lake

```
$ cd release_package/ubuntu/test_cmd/  
$ ./adl_gmsl_up_abcdefgh.sh
```



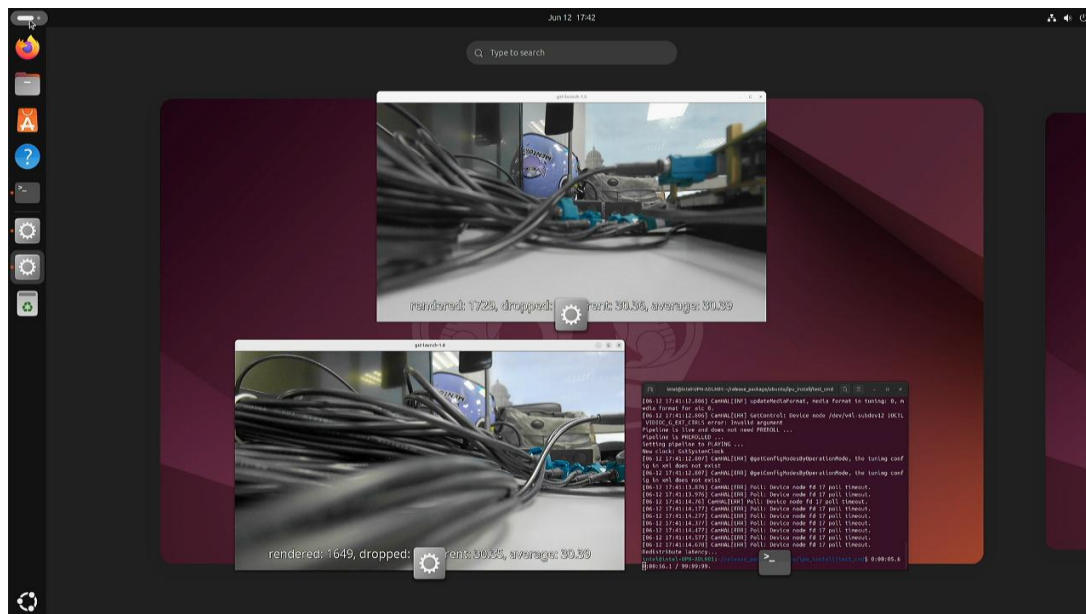
■ MIPI x 2

■ Meteor Lake

```
$ cd release_package/ubuntu/test_cmd/
$ ./mtl_mipi_up_ae.sh
```

■ **Alder Lake**

```
$ cd release_package/ubuntu/test_cmd/
$ ./adl_mipi_up_bc.sh
```



Camera support size list command

Camera support size can be listed based on **<Camera name>**.

```
$ v4l2-ctl -d $(media-ctl -e " <Camera name>") --list-subdev-framesizes code=0x200f
```

<Camera name> can be listed by [Camera list Command](#) mentioned above.

e.g.

```
$ v4l2-ctl -d $(media-ctl -e " APPRO_CAM a") --list-subdev-framesizes code=0x200f
ioctl: VIDIOC_SUBDEV_ENUM_FRAME_SIZE (pad=0,stream=0)
    Size Range: 1280x720 - 1280x720
    Size Range: 1920x1080 - 1920x1080
    Size Range: 2560x1440 - 2560x1440
    Size Range: 2560x1920 - 2560x1920
    Size Range: 3840x2160 - 3840x2160
```

Camera support framerate list command

Camera support framerate can be listed based on **<Camera name>** on different **<width>** and **<height>**.

```
$ v4l2-ctl -d $(media-ctl -e " <Camera name>") --list-subdev-frameintervals
width=<width>, height=<height>, code=0x200f
```

<Camera name> can be listed by [Camera list Command](#) mentioned above.

<width> and **<height>** can be listed by [Camera support size list command](#) mentioned above.

e.g.

```
$ v4l2-ctl -d $(media-ctl -e "APPRO_CAM a") --list-subdev-frameintervals width=1280,
height=720, code=0x200f
ioctl: VIDIOC_SUBDEV_ENUM_FRAME_INTERVAL (pad=0,stream=0)
    Interval: 0.200s (5.000 fps)
    Interval: 0.067s (15.000 fps)
    Interval: 0.040s (25.000 fps)
    Interval: 0.033s (30.000 fps)
```

Camera current framesize get command

Camera current framesize can get based on **<Camera name>**.

```
$ v4l2-ctl -d $(media-ctl -e "<Camera name>") --get-subdev-fmt
```

<Camera name> can be listed by [Camera list Command](#) mentioned above.

e.g.

```
$ v4l2-ctl -d $(media-ctl -e "APPRO_CAM a") --get-subdev-fmt
ioctl: VIDIOC_SUBDEV_G_FMT (pad=0,stream=0)
    Width/Height      : 1280/720
    Mediabus Code      : 0x200f (MEDIA_BUS_FMT_UYVY8_1X16)
    Field              : None
    Colourspace         : Default
    Transfer Function   : Default (maps to Rec. 709)
    YCbCr/HSV Encoding : Default (maps to ITU-R 601)
    Quantization        : Default (maps to Limited Range)
```

Camera current framerate get command

Camera current framerate based on **Current framesize** can get based on **<Camera name>**.

```
$ v4l2-ctl -d $(media-ctl -e "<Camera name>") --get-subdev-fps
```

<Camera name> can be listed by [Camera list Command](#) mentioned above.

Current framesize can get by [Camera current framesize command](#) mentioned above.

e.g.

```
$ v4l2-ctl -d $(media-ctl -e "APPRO_CAM a") --get-subdev-fps
ioctl: VIDIOC_SUBDEV_G_FRAME_INTERVAL (pad=0,stream=0)
    Frames per second: 30.000 (30/1)
```

Camera framerate set command

Camera framerate can be set based on **<Camera name>** on different **<width>** and **<height>**

NOTE:

framerate should be set before start preview. Don't set framerate when preview is running.

```
$ v4l2-ctl -d $(media-ctl -e " <Camera name>" ) --set-subdev-fmt=width=<width>, height=<height>  
$ v4l2-ctl -d $(media-ctl -e " <Camera name>" ) --set-subdev-fps fps=<fps>
```

<Camera name> can be listed by [Camera list Command](#) mentioned above.

<width> and **<height>** can be listed by [Camera support size list command](#) mentioned above.

<fps> can be listed by [Camera framerate list command](#) mentioned above.

e.g.

```
$ v4l2-ctl -d $(media-ctl -e "APPRO_CAM a") --set-subdev-fmt=width=1920, height=1080  
$ v4l2-ctl -d $(media-ctl -e "APPRO_CAM a") --set-subdev-fps fps=15
```

Camera image control list command

Camera image control interface can be listed based on **<Camera name>**

```
$ v4l2-ctl -d $(media-ctl -e "<Camera name>") -L
```

<Camera name> can be listed by [Camera list Command](#) mentioned above.

NOTE:

Different camera may have different control interface based on sensor's feature.

e.g.

```
$ v4l2-ctl -d $(media-ctl -e "APPRO_CAM a") -L
```

User Controls

```
brightness 0x00980900 (int) : min=0 max=255 step=1 default=128 value=128 flags=slider
contrast 0x00980901 (int) : min=0 max=255 step=1 default=128 value=128 flags=slider
saturation 0x00980902 (int) : min=0 max=255 step=1 default=128 value=128 flags=slider
hue 0x00980903 (int) : min=0 max=255 step=1 default=128 value=128 flags=slider
horizontal_flip 0x00980914 (bool) : default=0 value=0 flags=slider
vertical_flip 0x00980915 (bool) : default=0 value=0 flags=slider
power_line_frequency 0x00980918 (menu) : min=0 max=1 default=0 value=0 (60 Hz)
    0: 60 Hz
    1: 50 Hz
sharpness 0x0098091b (int) : min=0 max=255 step=1 default=128 value=128 flags=slider
denoise 0x00981b00 (int) : min=0 max=255 step=1 default=128 value=128 flags=slider
wdr_mode 0x00981b01 (menu) : min=0 max=2 default=1 value=1 (WDR)
    0: OFF
    1: WDR
    2: HDR
wdr_level 0x00981b02 (int) : min=1 max=10 step=1 default=3 value=3 flags=slider
digital_zoom 0x00981b03 (int) : min=0 max=31 step=1 default=0 value=0 flags=slider
pan 0x00981b04 (int) : min=1 max=100 step=1 default=50 value=50 flags=slider
tilt 0x00981b05 (int) : min=1 max=100 step=1 default=50 value=50 flags=slider
ae_lock 0x00981b06 (bool) : default=0 value=0 flags=slider
...
...
```

Camera image control set command

Camera image control interface **<ctrl>** can be set by **<val>** based on **<Camera name>**

```
$ v4l2-ctl -d $(media-ctl -e " <Camera name>") --set-ctrl <ctrl>=<val>
```

<Camera name> can be listed by [Camera list Command](#) mentioned above.

<ctrl> and range of **<val>** can be listed by [Camera image control list command](#) mentioned above.

```
$ v4l2-ctl -d $(media-ctl -e "APPRO_CAM a") --set-ctrl brightness=50
```

Camera image control get command

Current value of Camera image control interface **<ctrl>** can get based on **<Camera name>**

```
$ v4l2-ctl -d $(media-ctl -e " <Camera name>") --get-ctrl <ctrl>
```

<Camera name> can be listed by [Camera list Command](#) mentioned above.

<ctrl> can be listed by [Camera image control list command](#) mentioned above.

```
$ v4l2-ctl -d $(media-ctl -e "APPRO_CAM a") --get-ctrl brightness  
brightness: 128
```